NERC
NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

Draft White Paper: Model Portability for IBR EMT Studies

A Lightweight Common Information Model Profile

November 2025

RELIABILITY | RESILIENCE | SECURITY




3353 Peachtree Road NE
Suite 600, North Tower
Atlanta, GA 30326
404-446-2560 | www.nerc.com

Table of Contents

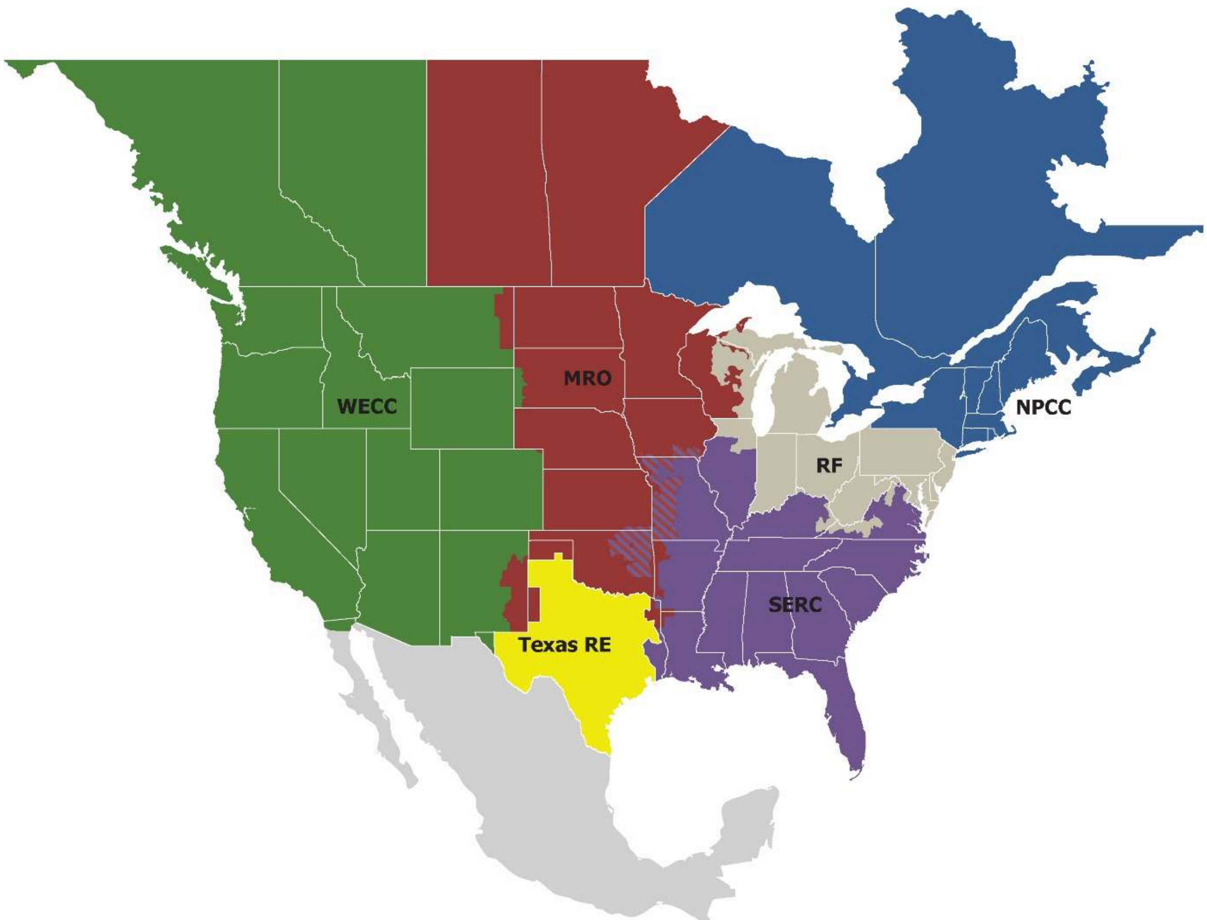
Preface

Electricity is a key component of the fabric of modern society and the Electric Reliability Organization (ERO) Enterprise serves to strengthen that fabric. The vision for the ERO Enterprise, which is comprised of NERC and the six Regional Entities, is a highly reliable, resilient, and secure North American bulk power system (BPS). Our mission is to assure the effective and efficient reduction of risks to the reliability and security of the grid.

Reliability | Resilience | Security

Because nearly 400 million citizens in North America are counting on us

The North American BPS is made up of six Regional Entities as shown on the map and in the corresponding table below. The multicolored area denotes overlap as some load-serving entities participate in one Regional Entity while associated Transmission Owners/Operators participate in another.



MRO	Midwest Reliability Organization
NPCC	Northeast Power Coordinating Council
RF	ReliabilityFirst
SERC	SERC Reliability Corporation
Texas RE	Texas Reliability Entity
WECC	WECC

Executive Summary

This white paper introduces the need for network model management in electromagnetic transient (EMT) studies. A standard data format would help users build, validate, update, and simulate EMT models to keep up with increasing demand for EMT studies of inverter-based resources (IBR). It also addresses key limitations of the IBR representations in existing formats by allowing for extensibility of model parameters and interfacing with external real-code models. A standard data format would also fulfill the intent of Federal Energy Regulatory Commission (FERC) Orders 901 and 2023 by improving model data sharing among Transmission Planners (TP) and Planning Coordinators (PC) for reliability studies even when they use different EMT software. The existing "base case" methods work for steady-state and dynamic network models in planning and operations, but these methods require customized, ad-hoc extensions for EMT modeling[1]. A recent NERC white paper has highlighted the importance of a "curated repository" of EMT network and IBR plant models, including real-code dynamic link libraries (DLL), to fill the gap in existing methods[2].

Three approaches were considered to solve this problem. The first approach would have extended and standardized "raw files," but this proved infeasible primarily for commercial reasons; it also entailed some technical shortcomings. The second approach extends the full International Electrotechnical Commission (IEC) standard Common Information Model (CIM), which has been successful in Europe with steady-state and dynamics modeling but has seen slow adoption in North America. The third approach adopts a simplified or lightweight CIM-inspired profile, optimized for EMT studies of IBRs, without meeting all possible CIM use cases. To be clear, the proposed EMT-IBR profile would not be fully CIM compliant but would leave the path open for upgrading to the full CIM. It includes an IEEE open-source software project with schema definitions, scripts, and test cases. The open-source project, and simplified CIM, should make the proposal more transparent and easier to adopt.

The roadmap for North American industry adoption would leverage IEEE standards, following this path:

- Complete the IEEE P3597 Standard for Interfacing Controls and Protection to Power System Simulation Tools[3]. This enables the "real-code DLLs"[4] to become part of the "curated repository." This standard is proposed for completion by the end of 2029, but the DLLs are already in use by EMT tool vendors and technology providers. See **Appendix C:** for one free-to-use and three open-source examples of the DLL interface. These examples provide a starting point for new implementers of the DLL interface.
- Complete the IEEE P3743 Recommended Practice for Electromagnetic Transient Model Interoperability for Electric Power Transmission Systems. This facilitates inputs from existing positive-sequence dynamics data to the EMT network model. It adds IBR balance-of-plant components and DLL parameter sets to the "curated repository" of IBR plant models. An open-source software project will be developed in parallel with the Recommended Practice. The method is based on a simplified or lightweight profile of the CIM, focused on EMT for IBR studies. Development of the Recommended Practice has been approved by the IEEE for completion by the end of 2028[5].
- Define a set of interoperability tests. For other IEC standards, (e.g., 61850 for intelligent electronic devices (IED) or 61968 and 61970 for the CIM), the interoperability tests have proven valuable in identifying gaps in

[1] J. Thapa et al., "Plug-and-Play Regional Models for Real-Time Electromagnetic Transient Simulations of Large-Scale Power Grids: A Case Study of New York State Power Grid," IEEE Access, vol. 11, pp. 87600–87614, 2023, doi: 10.1109/ACCESS.2023.3305394.

[2] NERC EMTWG, "Electromagnetic Transient Analysis in Operations Planning for BPS-Connected Inverter-Based Resources," June 2025

[3] IEEE Standards Association, "IEEE P3597, Standard for Interfacing of Controls and Protection to Power System Simulation Tools." Accessed: July 08, 2025. [Online]. Available: https://standards.ieee.org/ieee/3597/12053/

[4] A real-code DLL is one type of "black box" model. Black box models can be 1) real-code DLLs written to the Cigre/IEEE interface specification, or 2) hidden collections of library components for a specific EMT tool. Only the real-code DLL version of a black box model is interoperable among EMT tools.

[5] IEEE Standards Association, "IEEE P3743, Recommended Practice for Electromagnetic Transient Model Interoperability for Electric Power Transmission Systems." Accessed: Oct. 20, 2025. [Online]. Available: https://standards.ieee.org/ieee/3743/12233/

the standards. Those gaps are then addressed rapidly through the open-source software project for P3743. The interoperability tests will demonstrate to the EMT tool vendors and their users that P3597 and P3743 work correctly to implement the “curated repository.” The CIM user’s group might administer these tests, as they currently do for the IEC standards 61850, 61968, and 61970. Utility witnesses participate in these tests.

- NERC may then encourage utilities to adopt IEEE P3597 and P3743. For example, updates to or technical references for MOD-026-2, MOD-032-2, and MOD-033-3 could cite IEEE standards in application guidance as methods that would meet new NERC requirements for EMT models of IBRs. Other methods should be allowed, subject to the MOD-032 requirement “Data must be shareable on an interconnection-wide basis to support use in the interconnection-wide cases.” That encourages EMT tool vendors and IBR model providers to adopt P3597 and P3743 to guarantee shareability.

The main body of this white paper discusses the essential CIM features and extensions for EMT studies of IBR integration. Simplifications of the CIM are suggested to streamline its adoption in North America. There are some heuristics for extending positive-sequence data to include essential zero-sequence parameters for EMT. Negative sequence parameters are generally equal to positive sequence parameters, except for rotating machines, which have X2 approximately equal to the lowest machine transient reactance. The transformer winding connections should be modeled for their negative sequence phase shifts and effects on zero-sequence impedance.[6] One appendix presents the draft IEEE Standards Project Authorization Request for P3743. Another appendix documents one free-to-use and three open-source examples of the DLL interface, which can be reused by adopters of the new standard data format.

[6] EMT tools accept sequence parameter inputs for some component types. Internally, most EMT tools don’t solve sequence networks at run-time. For example, EMT tools may use phase impedance matrices, explicit neutral impedances and winding connections, custom modal decompositions for lines, dq0 decompositions for machines, etc.

Chapter 1: EMT Model Repository Requirements

As IBRs appear in greater numbers on the BPS, NERC has encouraged utilities to perform EMT studies of IBRs[7]. This creates a new burden on the planning engineers to create and maintain robust EMT models of many IBR plants, along with extensive BPS network models for EMT. Network models for power flow (PF) and positive-sequence phasor domain transient (PSPDT)[8] applications are maintained in tool-specific file formats that have become de facto standards. However, PF and PSPDT models do not have enough detail for EMT studies, so their de facto standard file formats are not sufficient for EMT. Protection engineers use short-circuit (SCKT) models to calculate fault current, but SCKT models and tools may not have acceptable accuracy for IBR studies.[9] In those cases, EMT models should be considered instead of SCKT tools to calculate fault currents in systems with IBR. As with all power system studies, the EMT model provides the foundation for all processes and results, and building the model can be the most time-consuming phase of an EMT study. This white paper summarizes approaches to reduce the effort and increase the quality of EMT model-building.

Currently, engineers must assemble, verify, and fill gaps in EMT model components manually, which is both time intensive and error prone. This process is semi-automated, based on PF/PSPDT files[10], with specific gaps in zero-sequence data, geographic information, transformer and bus mismatches, boundary identification and equivalencing, validating solutions, dynamic load modeling, and synchronizing updates[11]. A complete EMT model includes the transmission network, loads, and generator plants, along with IBR plants.

Instead of storing links to an EMT model of IBR plants, one could store links to a standardized model. This approach has shown success with CIM-based EMT model building in France[12] and in Europe for HVDC[13]. Proprietary solutions have been used in Japan[14]. The *CIM Network and Dynamics* schema[15,16,17] can support a framework for building and maintaining EMT models for IBR studies.

Text file formats (TFF), sometimes called “raw files,” were initially considered to implement an EMT repository. The TFF would require extensions for EMT. The most widely used TFF in North America was developed by a simulation tool vendor that maintains control of that format. For a new and broader use case, like EMT network modeling, the

---

[7] NERC, “Reliability Guideline: Electromagnetic Transient Modeling for BPS Connected Inverter-Based Resources—Recommended Model Requirements and Verification Practices,” North American Electric Reliability Corporation, Atlanta, Mar. 2023. [Online]. Available: https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Reliability_Guideline-EMT_Modeling_and_Simulations.pdf

[8] A more precise term for “stability” analysis, under consideration by the NERC EMT Task Force and the IEEE P2800.2 Working Group.

[9] Z. Cheng, T. Patel, J. Holbach, and M. Reno, “Assessing Inverter-Based Resources Modeling Gaps in Commonly Used Short-Circuit Programs,” SAND--2024-13494, Oct. 2024. doi:10.2172/2480122

[10] P. Zhao et al., “Rapid Modeling of Large-Scale Power System Based on PSCAD,” in 2019 IEEE Asia Power and Energy Engineering Conference (APEEC), Mar. 2019, pp. 167–171. doi: 10.1109/APEEC.2019.8720692.

[11] M. Abdelmalak et al., “PSS/E to RSCAD Model Conversion for Large Power Grids: Challenges and Solutions,” in 2021 IEEE Power & Energy Society General Meeting (PESGM), July 2021, pp. 01–05. doi: 10.1109/PESGM46819.2021.9637930.

[12] F. Martin, C. Y., “Automation of model exchange between planning and EMT tools,” presented at the International Conference on Power System Transients (IPST), Seoul, 2017. Accessed: Dec. 20, 2023. [Online]. Available: https://www.ipstconf.org/papers/Proc_IPST2017/17IPST099.pdf

[13] C. Ivanov, G. Tishenin, D. Lanzarotto, F. Morel, and A. Monti, “Standardized Model Exchange of HVDC Equipment for RMS and EMT Simulations,” in 2024 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE), Oct. 2024, pp. 1–5. doi: 10.1109/ISGTEUROPE62998.2024.10863427.

[14] T. Noda, T. Tadokoro, and T. Dozaki, “Automatic Generation of Power System Simulation Data Cases From Utility Databases: Introducing a new technology,” IEEE Electrification Magazine, vol. 11, no. 4, pp. 79–85, Dec. 2023, doi: 10.1109/MELE.2023.3320521.

[15] CIM Users Group, “Standards Artifacts - CIM Users Group.” Accessed: July 08, 2025. [Online]. Available: https://cimug.org/cimdocs/standards-artifacts/

[16] ENTSO-E, “Common Grid Model Exchange Standard (CGMES) Library.” Accessed: Jan. 31, 2025. [Online]. Available: https://www.entsoe.eu/data/cim/cim-for-grid-models-exchange/

[17] Another link at https://cgmes.github.io/dynamics/#dynamics was taken down late in 2024, possibly due to intellectual property concerns of IEC and/or ENTSO-E.

CIM seems more appropriate for standardization. The CIM is agnostic to different vendor file formats. The CIM also supports vendor- and utility-specific extensions.

The CIM is the subject of two families of IEC standards, 61970 and 61968. A fully compliant IEC CIM was considered for EMT modeling and partly developed[18]. However, the complete CIM standard is probably too complicated for a rapid adoption in North America for EMT IBR modeling. Instead, a simpler CIM-inspired EMT-IBR profile is proposed in this white paper.[19]

Figure 1.1 shows the framework to implement a repository of EMT network and IBR plant models. The network model is usually based on bus-branch PSPDT and SCKT data. The *Other* data may include protective device settings, node-breaker topologies, and extensions to be determined later. EMT simulators also require *Initial Conditions* to begin time stepping, so the network model should support different sets of steady-state solutions to provide these initial conditions. In the utility workflow, this network model may be updated at three-to-six-month intervals. There may also be different versions of the network model for different planning scenarios. Under different guises, the CIM already supports PSPDT, SCKT, bus-branch, node-breaker, initial conditions, and planning scenarios.

However, the CIM does not currently support a repository of IBR models. The utility maintains its own network model, but various project developers and hardware vendors provide most of the IBR data. The IBR models include custom parameters, custom logic, and, increasingly, custom binary control code in the form of a DLL[20]. Some of the PSPDT tools support these in the form of "user code" but not in a standardized way and not in a way that supports EMT. As with the network model, IBR models must be maintained in different versions and validated.

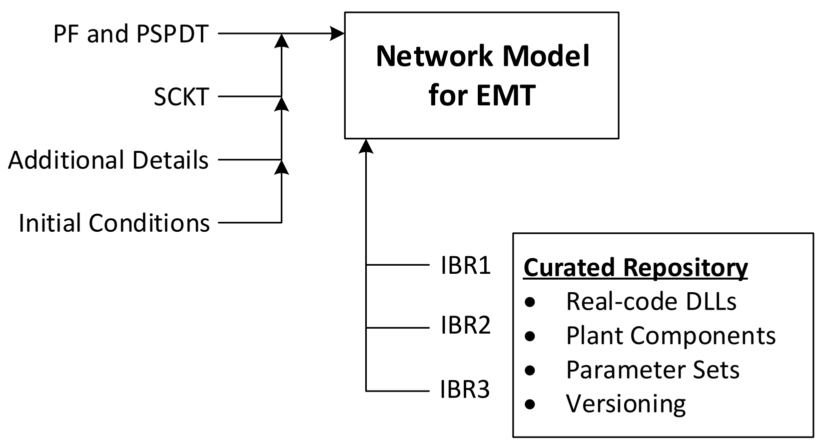


Figure 1.1: Attributes of an EMT Network and IBR Model Repository [2]

Landscape for Adoption

The path for NERC adoption of the EMT network repository in Figure 1.1 would leverage IEEE standards. Recently, IEEE 2800-2022 has helped NERC adopt performance requirements for BPS-connected IBRs. Some lessons have been learned from that process. For the EMT network repository, these include the following:

[18] T. E. McDermott, "Common Information Model for Electromagnetic Transients (CIM for EMT): CRADA 533 [Abstract only}," Pacific Northwest National Lab. (PNNL), Richland, WA (United States); 9327-6806 Quebec Inc., Montreal, QC (Canada), PNNL-32479, Jan. 2022. doi: 10.2172/1867325.

[19] The starting point may be CIM_Grid18v15_Enterprise14v04_Market04v18, which is downloadable from the CIM user's group web site. It includes the WECC generic IBR models and aggregate DER models from IEC 61970:302-2024.

[20] Cigre JWG B4.82/IEEE, "Guidelines for use of real-code in EMT models for HVDC, FACTs and inverter based generators in power systems analysis," Cigre, Paris, France, Technical Brochure 958, Feb. 2025. [Online]. Available: https://www.e-cigre.org/publications/detail/958-guidelines-for-use-of-real-code-in-emt-models-for-hvdc-facts-and-inverter-based-generators-in-power-systems-analysis.html

- Complete the IEEE P3597 Standard for Interfacing Controls and Protection to Power System Simulation Tools. This enables the "real-code DLLs" to become part of the "curated repository." The standard is proposed for completion by the end of 2029, but the DLLs are already in use by EMT tool vendors and technology providers.
- Complete the IEEE P3743 Recommended Practice for Electromagnetic Transient Model Interoperability for Electric Power Transmission Systems. This enables PF, PSPDT, and SCKT inputs to the EMT network model. It adds balance-of-plant components and DLL parameter sets to the "curated repository" of IBR plant models. An open-source software project will be developed in parallel with the Recommended Practice. The method is based on a simplified or lightweight profile of the CIM, focused on EMT for IBR studies. Development of the Recommended Practice has been approved by to IEEE for completion by the end of 2028.
- Define a set of interoperability tests, as illustrated in **Figure 1.2**. For other IEC standards, (e.g., 61850 for IEDs or 61968 and 61970 for the CIM), the interoperability tests have proven valuable in identifying gaps in the standards. Those gaps are then addressed rapidly through the open-source software project for P3743. The interoperability tests will demonstrate to the EMT tool vendors and their users that P3597 and P3743 work correctly to implement the EMT model repository. The CIM user's group might administer these tests, as they currently do for the IEC standards 61850, 61968, and 61970. Utility witnesses participate in these tests.
- NERC may then encourage utilities to adopt IEEE P3597 and P3743. For example, an update to MOD-032-2 could cite IEEE standards in application guidance as methods that would meet new NERC requirements for EMT models of IBRs. Other methods should be allowed, subject to the MOD-032 requirement "Data must be shareable on an interconnection-wide basis to support use in the interconnection-wide cases." That encourages EMT tool vendors and IBR model providers to adopt P3597 and P3743 to guarantee shareability.

The simulation tools referenced in **Figure 1.2** include the following:

- **Raw Files:** The legacy text input files used for PF, PSPDT, and sometimes SCKT analysis. The CIM input interface is partly implemented. However, the raw file schema is not publicly available, so users of it may be reverse engineering at their own risk. The CIM output interface should be implemented to support round-trip interoperability testing, which can pinpoint defects in either the CIM schema or conversion scripts.
- **OpenDSS:** An open-source simulator of unbalanced (non-transposed) distribution systems[21], used for studies of distributed energy resources (DER), including time-series power flow and dynamics[22]. The CIM input and output interfaces are fully implemented. It is included here to show 1) the ability of the CIM to model unbalanced systems and 2) the importance of round-trip interoperability testing for CIM inputs and outputs.
- **MATPOWER:** An open-source optimal power flow tool[23], used for teaching and research. The CIM interface is already implemented. It is included here to demonstrate the feasibility of PF output from the CIM to a different tool than used for the input source. MATPOWER can then provide initial conditions for EMT.
- **Alternative Transients Program** (ATP): A free-to-use EMT tool[24], not widely used for IBR studies in North America. The CIM interface is already implemented. It is included here to show the ability of the CIM to model transmission systems with IBR and other generator dynamics.
- **Other EMT** – Includes other CIM-to-EMT interfaces to be implemented by the tool providers. A small number of these may already be implemented. Interfaces for open-source[25] and free-to-use[26] EMT tools could also be implemented and published to serve as examples.[27]

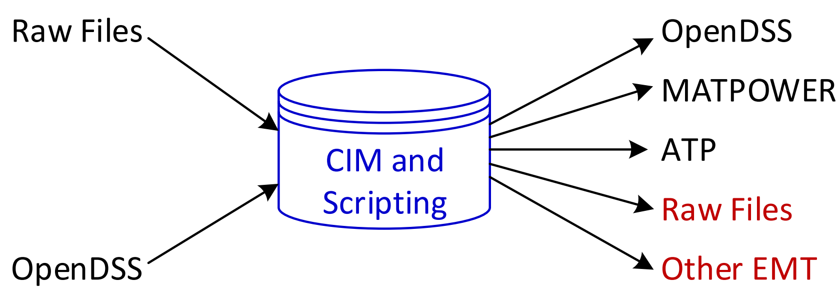


Figure 1.2: Implementation of CIM Input and Output (Items TO-DO in Red)

Referencing the Standard Data Format in NERC Standards

NERC establishes and maintains grid reliability standards independent of the IEEE, the IEC, and other standards development organizations. In the Unites States, NERC's authorization to do this comes from FERC[28]. For example, FERC Order 901 directs NERC to develop new Reliability Standards for IBRs that address gaps in data sharing, model

[21] R. C. Dugan and T. E. McDermott, "An open source platform for collaborating on smart grid research," in 2011 IEEE Power and Energy Society General Meeting, July 2011, pp. 1–7. doi: 10.1109/PES.2011.6039829.

[22] EPRI, "OpenDSS Dynamics Mode," OpenDSS Documentation. Accessed: July 08, 2025. [Online]. Available: https://opendss.epri.com/OpenDSSDynamicsMode.html

[23] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, "MATPOWER: Steady-State Operations, Planning, and Analysis Tools for Power Systems Research and Education," IEEE Transactions on Power Systems, vol. 26, no. 1, pp. 12–19, Feb. 2011, doi: 10.1109/TPWRS.2010.2051168.

[24] EEUG, "Alternative Transients Program." Accessed: July 08, 2025. [Online]. Available: https://www.atp-emtp.org/

[25] M. Xiong et al., "ParaEMT: An Open Source, Parallelizable, and HPC-Compatible EMT Simulator for Large-Scale IBR-Rich Power Grids," IEEE Transactions on Power Delivery, vol. 39, no. 2, pp. 911–921, Apr. 2024, doi: 10.1109/TPWRD.2023.3342715.

[26] D. Daigle, "Kestrel EMT – Electromagnetic transient simulation for free." Accessed: April 21, 2026. [Online]. Available: https://arnadaeng.com/kestrel-emt

[27] Add https://www.ornl.gov/sites/default/files/2025-09/RE_INTEGRATE_DAE_Solvers.pdf after it appears in IEEExplore.

[28] NERC also works with grid regulators in Canada and Mexico.

validation, planning and operational studies, and IBR performance requirements[29]. NERC is partway through the process of meeting this FERC directive. Some of the points made in FERC Order 901 include the following:

- Studies must assess performance of all IBRs across planning and operational boundaries (p. 8). This cannot be achieved without better methods of building and maintaining network and IBR models for EMT and PSPDT.
- IEEE and other standardization efforts might be inconsistent or incomplete and they do not have comprehensive plans for adoption and imposing mandatory requirements (p. 22). This partially explains why NERC does not rely on IEEE standards. Furthermore, IEEE and other standards may change outside of NERC's control.
- Multiple commenters asked FERC to require EMT modeling. FERC declined to do so but encouraged NERC to keep working on EMT modeling (p. 90). This justifies NERC incorporating EMT requirements, especially for cases where PSPDT simulations fail to predict IBR behavior. The order's assumption that "EMT models are not used to build interconnection-wide models" may already be outdated.
- FERC insisted on generic IBR models, characterized as "nationwide approved" models, in part to assure robust simulation in a variety of PSPDT tools. The user-code models were described as useful for interconnection studies but not essential for large-scale studies, also prone to numerical problems that are hard to debug (p. 90–95). That assumption may already be outdated by recent advances, like the DLL interface specification. Furthermore, FERC suggested that NERC may continue to <u>allow</u> user-defined models to be submitted along with the approved (and required) generic models.

A recent NERC protection and control (PRC) standard[30] makes reference to IEEE standards[31,32] as methods of meeting NERC's requirements on disturbance monitoring data exchange. However, the IEEE standard that would be most burdensome for new implementers is not mandatory because NERC also allows comma-separated value (CSV) files "with appropriate headers." CSV files have some drawbacks that put a burden on the recipients, who will probably convert them to the IEEE format anyway after translating the "appropriate headers." But using the CSV files reduces the burden on parties who must provide the files.[33] In this case, NERC leverages the extensive work done on the IEEE Common Format for Transient Data Exchange (COMTRADE) for Power Systems format over several decades without having to redo the work. Equally important, stakeholders can already acquire many instruments, software products, and even open-source software tools that support COMTRADE.

NERC has three modeling, data, and analysis (MOD) standards that might be updated for EMT IBR model interoperability. MOD-032-2 addresses the network model maintenance, including updates for IBRs now in final balloting[34]. In the preamble to Attachment 1, MOD-032-2 already states, *"Data must be shareable on an interconnection-wide basis to support use in the interconnection-wide cases."* Shareability requires that models must be portable among stakeholders, and among simulation tools (i.e., vendor lock-in should be avoided).

[29] FERC, Reliability Standards to Address Inverter-Based Resources, Oct. 23, 2023. Accessed: Aug. 04, 2025. [Online]. Available: https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20231019-3157

[30] NERC, PRC-002-5 – Disturbance Monitoring and Reporting Requirements, Feb. 20, 2025. [Online]. Available: https://www.nerc.com/pa/Stand/Reliability%20Standards/PRC-002-5.pdf

[31] IEEE, C37.111-2013 - IEEE/IEC Measuring relays and protection equipment – Part 24: Common format for transient data exchange (COMTRADE) for power systems, Apr. 2013. doi: 10.1109/IEEESTD.2013.6512503.

[32] IEEE, C37.232-2011 - IEEE Standard for Common Format for Naming Time Sequence Data Files (COMNAME), Nov. 2011. doi: 10.1109/IEEESTD.2011.6081885.

[33] When this report was written, the cost of [24] is $90 to non-members of IEEE. Data files must be named according to that standard for PRC-002-5. The cost of [23] is $131 to non-members of IEEE, but it is not required when submitting CSV files. Most users won't need to purchase either standard because their software already supports them, or the users can follow documented examples of proper file naming.

[34] NERC, MOD-032-2 - Data for Power System Modeling and Analysis, Aug. 08, 2025. [Online]. Available: https://www.nerc.com/pa/Stand/Project202202ModificationstoTPL00151andMOD0321DL/2022-02%20MOD-032-2_Clean_August_8_2025.pdf

The NERC CIP-13 standards require a supply chain cyber security risk management plan that includes transitions from one vendor to another vendor. CIP-13 applies to Bulk Electric System Cyber Systems. This white paper proposes extending the same idea to EMT planning software models. Vendor lock-in to EMT tools poses a supply chain risk, by increasing the cost, time and risk of errors in a transition from one vendor to another vendor. EMT vendors might go out of business, fall behind in technology development, or pose other business risks. Vendor lock-in also impedes the practical shareability of modeling data. Shareability includes permission to use modeling data. It also requires technical shareability, i.e., the different EMT tools must be able to read and execute on the modeling data.

So far, power system model data for steady-state, dynamics, and short circuit has been shareable under MOD-032-2 via vendor-controlled TFF files. These TFF files have been reverse-engineered by other software vendors and stakeholders. There are some EMT tools with proprietary file formats, but they are not publicly documented and not suitable for sharing with users of other EMT tools. A proprietary EMT model management tool could still meet NERC shareability requirements if it could export data, without loss or error, into a publicly documented format, thereby avoiding vendor lock-in.

These updates are proposed for MOD-032-2:

- In Requirement R1 of the Reliability Standard, Part 1.4, change to “Specifications of the following items for dynamic **and EMT** model submissions…”
- In Requirement R2 of the Reliability Standard, Part 2.2, change to “If a functional entity is required to provide dynamic **or EMT** models…”
- There are about 19 uses of the phrasing “steady-state, dynamics, and short circuit” in the Reliability Standard. Most of these are in the Table of Violation Severity Levels, but one is in R1 and one is in R2. Replace each occurrence with “steady-state, dynamics, short circuit**, and EMT**.”
- In the table of Attachment 1, **dynamics** column, insert “7b. Identification of components that are part of an IBR plant [GO].”
- In the table of Attachment 1, add a fourth column for **electromagnetic transients** with the following items:
  - Include all applicable elements in columns “steady-state,” “dynamics,” and “short circuit”
  - Real-code DLL models for IBR. These may also run in dynamics [GO]
  - Transformer saturation, either typical or from test reports [GO, TO]
  - Line conductors and geometry are preferred to Mutual Line Impedance Data for EMT [TO]
- In the Technical Rationale for Requirement R1, add some EMT-related information: “The NERC Inverter-Based Resource Subcommittee has recently published a Reliability Guideline: Electromagnetic Transient Modeling for BPS-Connected Inverter-Based Resources. This guideline identifies situations that may call for electromagnetic transient (EMT) modeling, which requires data beyond the data required for steady-state, dynamics, and short circuit modeling.”
- In the Technical Rationale for Requirement R1, Part 1.4.2, add, “Real-code dynamic link library (DLL) models meet this requirement if they follow, or the latest version of IEEE Standard 3597.”
- To the Technical Rationale for Attachment 1, insert a paragraph suggesting use of IEEE standards to meet R1. “Model data must be shareable on an interconnection-wide basis to support interconnection-wide cases as required by MOD-32-2, and to avoid vendor lock-in. At a minimum, data formats must be documented and accessible. Standard formats would further reduce the cost and error rate of data conversions. To help achieve these goals, stakeholders may use IEEE Standards 3597 and 3743 as they are developed, including their companion open-source software. These publicly define standard formats for power system data. Other methods and systems may be used, so long as those alternatives can still provide all data in a shareable

format. IEEE Standards 3597 and 3743 focus on the evolving needs in EMT data, but also encompass steady-state, dynamics, and short circuit data."

MOD-033-3 addresses the network model validation[35], including updates for IBRs that passed final balloting. These further updates, which broaden the scope of dynamics validation to include EMT when needed, are proposed:

- In requirement R1 of the Reliability Standard, point 1.2, replace "dynamic planning System model" with "dynamic or **EMT** planning System model." This flexibility may be important in cases where PSPDT simulation does not perform adequately for IBR events.
- In the Technical Rationale for R1 overall, pages 1–2:
    - Edit the paragraph that begins "Implementation of these validation requirements…": "Implementation of these validation requirements will result in more accurate power flow**, dynamic, and electromagnetic transient (EMT)** system models. This, in turn, should result in better correlation between system flows and voltages seen on power flow studies and the actual values occurring in system operations. Similar improvements should be expected for dynamics **and EMT** studies, such that the result will more closely match the actual responses of the power system to disturbances."
    - Append to the next paragraph that begins "Requirement R1 focuses…": "A recent NERC Reliability Guideline: Electromagnetic Transient Modeling for BPS Connected Inverter-Based Resources—Recommended Model Requirements and Verification Practices has identified the need for EMT simulation in some cases where dynamic simulation does not match the performance of IBRs."
- In the Technical Rationale for R1, Part 1.2, bottom of page 3, begin the paragraph with: "The scope of dynamics **and EMT** model validation in Requirement 1, Part 1.2 …"
- In the Technical Rationale for R1, Part 1.3, page 6, after the discussion of visual plot inspection, reference IEEE 2800 and P2800.2 for guidance on model validation for IBR.
- In the Technical Rationale for R1, Part 1.4, page 6: "If a generic IBR model does not match the actual response, then a real-code DLL model (for dynamics and/or EMT) should be requested from the data provider."
- MOD-026-2 addresses dynamic model validation for generators, including updates for HVdc, flexible ac transmission systems (FACTS), and IBRs that passed final balloting[36]. The updated requirement R1 calls for Transmission Planners (TP) and Planning Coordinators (PC) to specify acceptable EMT models and formats as necessary for IBRs. No updates are proposed for the Reliability Standard, but these updates are proposed for its technical rationale:
    - Under Part 1.2, add a line item: "8. For accuracy, interoperability, and consistency, it is recommended that the TP requires use of IEEE Standard 3597 to implement real-code models of IBR controllers in dynamics and EMT tools."
    - Under Part 1.2, add a line item: "9: For interoperability and consistency, it is recommended that the TP requires use of IEEE Recommended Practice 3743 to format IBR plant data for dynamics and EMT."
    - Under Part 1.3, some of the line items may refer to a particular EMT tool. To discourage vendor lock-in, broader language should be used. For example, instead of the reference to Fortran, substitute, "The version of any compilers required for it to run."

[35] NERC, MOD-033-3 - Steady-State and Dynamic System Model Validation, Aug. 08, 2025. [Online]. Available: https://www.nerc.com/pa/Stand/Project202101_Modifications_to_MOD025_and_PRC019DL/2021-01%20Draft%202%20MOD-033-3%20clean%2008082025.pdf

[36] NERC, MOD-026-2 - Verification and Validation of Dynamic Models and Data, Aug. 14, 2025. [Online]. Available: https://www.nerc.com/pa/Stand/Project_2020_06_Verifications_of_Models_and_Data_f/2020-06%20MOD-026-2%20Clean%20Additional%20Ballot%201_08142025.pdf

- Under Part 1.3, add a bullet item: “Pre-compiled IBR control code in the format of IEEE Standard 3597.”
- Under Part 1.3, add a bullet item: “IBR plant model component data in the format of IEEE Recommended Practice 3743.”

Proof of Concept

Simulation results from a preliminary CIM-based framework have been obtained using the free-to-use alternative transients program (ATP) with two medium-scale BPS networks[37]:

- **IEEE118** – Representative of the Midwestern United States circa 1962, with 56 machines aggregating steam and hydro generators, 14 solar generators, and 5 wind generators. Includes 193 buses considering generator step-up (GSU) transformers[38].
- **WECC240** – Representative of the Western Electricity Coordinating Council (WECC) system, with 111 machines aggregating steam and hydro generators, 25 solar generators, and 10 wind generators. Includes nine series capacitors, two aggregations of DERs, and 243 buses[39].

Data and script files for these examples are available for ATP[40,41]. These examples also support MATPOWER, an open-source PF and optimal PF tool. These examples show how P3743 might be implemented in an open-source project. See **Appendix C:** for an example of how P3597 might be implemented in open source.

Long-Term Maintenance

The document of IEEE P3743 should be updated or withdrawn within 10 years after each new version. The companion open-source software site of P3743, which contains the electronic files and on-line documentation to implement the recommended practice, should be updated more frequently than ten years. Users and implementers of P3743 may also extend it locally. This allows the EMT repository schema to grow and respond quickly to evolving grid requirements. The document will describe processes and requirements for using and maintaining P3743, citing the open-source software site as a normative reference. The open-source software site will maintain the EMT repository schema under version control. The site will have a team of maintainers who review schema update requests and approve, revise, or deny those requests through online collaboration. The working group developing P3743 will identify a team of at least two, and preferably up to six, maintainers for the open-source site. Anyone can submit defect reports, feature requests, questions, and other changes through the “issues” forum on the site. Developers can submit proposed changes as “pull requests,” which the site administrators will test and review for deciding whether to merge the proposed change. Old versions of the schema will remain available for legacy applications. The open-source site is governed by policies and procedures of the IEEE Standards Association and Open-Source Software Committee.

The separate structure of document and open-source software site enables future growth:

---

[37] M. Ramesh and McDermott, Thomas E., “Transmission System Transient Models from the CIM,” presented at the IEEE PES General Meeting, Seattle, Washington, July 24, 2024. Accessed: Sept. 25, 2024. [Online]. Available: https://emthub.readthedocs.io/en/latest/_static/24PESGM2411.pdf

[38] I. Peña, C. B. Martinez-Anido, and B.-M. Hodge, “An Extended IEEE 118-Bus Test System With High Renewable Penetration,” IEEE Transactions on Power Systems, vol. 33, no. 1, pp. 281–289, Jan. 2018, doi: 10.1109/TPWRS.2017.2695963.

[39] H. Yuan, R. S. Biswas, J. Tan, and Y. Zhang, “Developing a Reduced 240-Bus WECC Dynamic Model for Frequency Response Study of High Renewable Integration,” Piscataway, NJ: Institute of Electrical and Electronics Engineers (IEEE), NREL/CP-5D00-79181, Jan. 2021. doi: 10.1109/TD39804.2020.9299666.

[40] T. McDermott, temcdrm/emthub. (Sept. 25, 2024). Python. Accessed: Sept. 25, 2024. [Online]. Available: https://github.com/temcdrm/emthub

[41] ATP is free to use, but not open source. The licensing terms prohibit use by organizations that have engaged in “EMT commerce” (e.g., EMT tool vendors and the Electric Power Research Institute). Most utilities and other organizations can obtain ATP licenses. The example input data and scripts for ATP in [32] are publicly available.

- Some CIM extensions from IEEE P3743 may be proposed to the IEC for adoption into the full standard (i.e., IEC 61970-301 and 61970-302). Approvals typically take years in the IEC process and require a majority vote of participating countries. Even if the extensions are not approved, they can still be used in North America under IEEE P3743. If a CIM extension is primarily made to enable rapid adoption in North America, it would not be proposed to the IEC.
- Some users and implementers may wish to adopt a more complete version of the CIM. IEEE P3743 will recommend an upgrade path into full CIM compliance, (e.g., by adopting the Common Grid Model Exchange Specification (CGMES)). The CGMES is already successful for dynamics model exchange in Europe. The full CIM also supports EMT modeling from physical device data. For example, transmission lines that share a right-of-way can be modeled using conductor and tower data and transformers can be modeled from manufacturer test reports.
- The DLL interface is a much-simplified version of the Functional Mockup Interface (FMI), which has been used in many industries like automative and aerospace but less so for electric power systems[42]. The FMI is more robust with more features than the DLL interface but is also more complicated. An IBR vendor or EMT tool vendor that already supports the DLL interface could later support the FMI interface with a relatively simple software adapter. The full CIM already includes links to the FMI, so an adopter of P3743 could add FMI support later.
- The EMT repository could also support PF, SCKT, and PSPDT data requirements. P3743 was not scoped for those use cases because they fall under the jurisdiction of other IEEE committees, such as Power System Dynamic Performance (PSDP). Co-sponsorships may be sought to expand the scope of P3743.
- Protection devices may contain hundreds of setting parameters and proprietary algorithms (i.e., they pose modeling challenges like those from IBRs). Many utilities keep protection device settings in the database of an SCKT tool, and those settings could be important in EMT simulations. New work on the full CIM includes protection device modeling, which may be adopted in North America. Alternatively, the DLL interface may be applied to protection devices in a later version of P3743's open-source software site.
- Real-time EMT simulators require custom logic blocks to complete a network model beyond the component and IBR data envisioned for the first version of P3743. So far, no basis of standardization has been identified for those real-time logic blocks, but it could be added later to the P3743 open-source software site.
- Vendors or utilities may wish to add custom extensions to the CIM, as CIM users. Both the simplified version in P3743 and the full version in IEC/CGMES allow this. Data and schemas may be added as a CIM user extension. CIM user extensions are designed not to interfere with the core CIM. Of course, they will be ignored by applications that do not know how to process the extended data. Examples might include custom EMT tool configurations for large-scale simulation and custom circuit breaker modeling with arc physics, pre-insertion resistors, point-on-wave switching, pole spans, etc. Some of these CIM user extensions may be proposed to IEEE P3743 through a "pull request" or to the full CIM through the IEC Working Groups, which promotes interoperability. However, there is no requirement to do so.

Participants in the P3743 Working Group, and the open-source software site maintainer team, need not become IEEE members. However, they would need to create free IEEE web accounts and accept some IEEE privacy policies.

[42] M. Cai, J. Mahseredjian, U. Karaagac, A. El-Akoum, and X. Fu, "Functional Mock-Up Interface Based Parallel Multistep Approach With Signal Correction for Electromagnetic Transients Simulations," IEEE Transactions on Power Systems, vol. 34, no. 3, pp. 2482–2484, May 2019, doi: 10.1109/TPWRS.2019.2902740.

Chapter 2: IBR Plant Model Components

This chapter discusses the difference between conventional generator plant models with rotating machines and IBR plant models with power electronic converters. The IBR plant models are more complicated because the plants are newer with less time for standardization to develop among model architectures and parameter sets. The IBR balance-of-plant models also typically contain more components than conventional generator plants.

Figure 2.1 shows a conventional rotating machine connected to a single-machine infinite bus (SMIB) test system at extra-high voltage (EHV). The plant is connected at a point of connection (POC). The generator model in *dq* axis form, a single inertia, *J*,[43] and typical control systems are highlighted in orange. Most of these control system models are built into the PSPDT library, with standardized parameter sets. PSPDT models, and the CIM for Dynamics schema, include these items. There is some usage of user-code models, which are not standardized among PSPDT tools and already create some model portability issues in dynamic studies.

The plant also minimally includes a GSU transformer, highlighted in green. Many balance-of-plant models only include a GSU, which only serves one generator. In such cases, there has been little need to identify the balance-of-plant components for special handling. The GSU data is generally available in PSPDT models, but the delta winding connection should be specified for EMT. The CIM already supports the items in **Figure 2.1**[44], partially so for the user-code models.

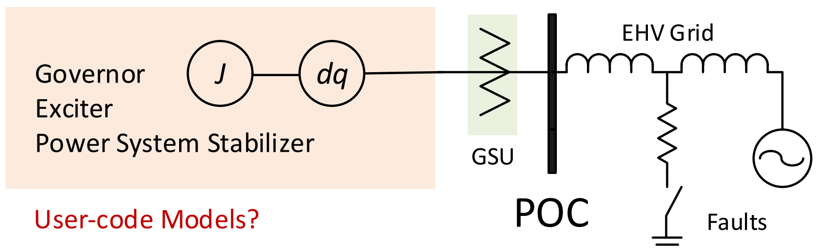


Figure 2.1: Rotating Machine Plant Model Components Connected to a SMIB Test System

Figure 2.2 shows an IBR plant model connected to the SMIB at a Reference Point of Applicability (RPA). The balance-of-plant model, highlighted in green, includes more components than in **Figure 2.1**, but they are all supported in the CIM. The IBR-specific components, highlighted in orange, are not presently supported in the CIM. Extensions to enable this support will be discussed next.

The *Exact or Generic Controls* in **Figure 2.2** may comprise a real-code DLL as defined in [14], with examples from EPRI[45,46]. It may also comprise a WECC generic model, which is already supported in CIM. Other options include custom user models, which would be specific to the EMT tool and not addressed in this white paper. **Figure 2.3** summarizes the DLL interface for one example. It supports a flexible number of parameters, input signals, and output signals. These extra parameters and signals include support for DC bus modeling. The DLL interface may be

[43] One of the early EMT applications was to perform sub-synchronous resonance (SSR) studies of generators in series-compensated BPS. For SSR studies, the single inertia, *J*, had to be replaced with a multi-mass rotational system. That data was not available in PSPDT models, which is still usually the case.

[44] PNNL, "i2x/emt-bootcamp at master · pnnl/i2x," GitHub. Accessed: Dec. 20, 2023. [Online]. Available: https://github.com/pnnl/i2x/tree/master/emt-bootcamp

[45] EPRI, "Code Based Generic Inverter Based Resource Model." Accessed: Sept. 25, 2024. [Online]. Available: https://www.epri.com/research/products/3002028322

[46] J. Boemer, W. Baker, and D. Ramasubramanian, "Generic Photovoltaic Inverter Model in an Electromagnetic Transients Simulator for Transmission Connected Plants: PV-MOD Milestone 2.7.3," June 2022. [Online]. Available: https://www.epri.com/pvmod

incorporated into the CIM by reference, especially after the DLL interface is formalized as an IEEE standard.[47] The orange-highlighted converter bridge and passive filter components in **Figure 2.2** might be specified in a tractable number of attributes for a new CIM class, as discussed later. EMT tools generally have their own inverter bridge modeling options, including average sources for either exact or generic controls, or pulse-width-modulated (PWM) switching topologies for exact controls.[48]

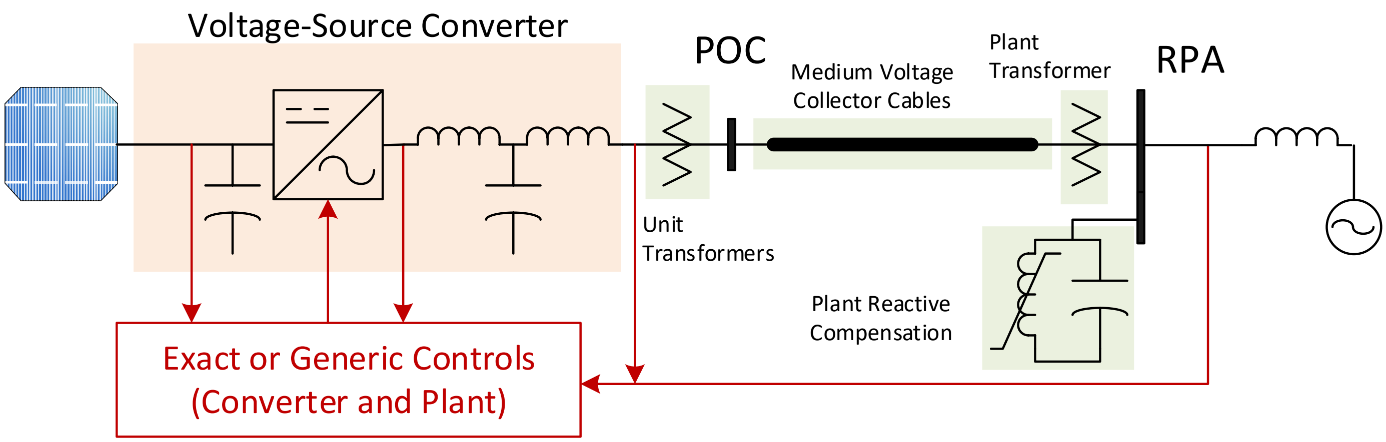


Figure 2.2: IBR Plant Model Components Connected to a SMIB Test System

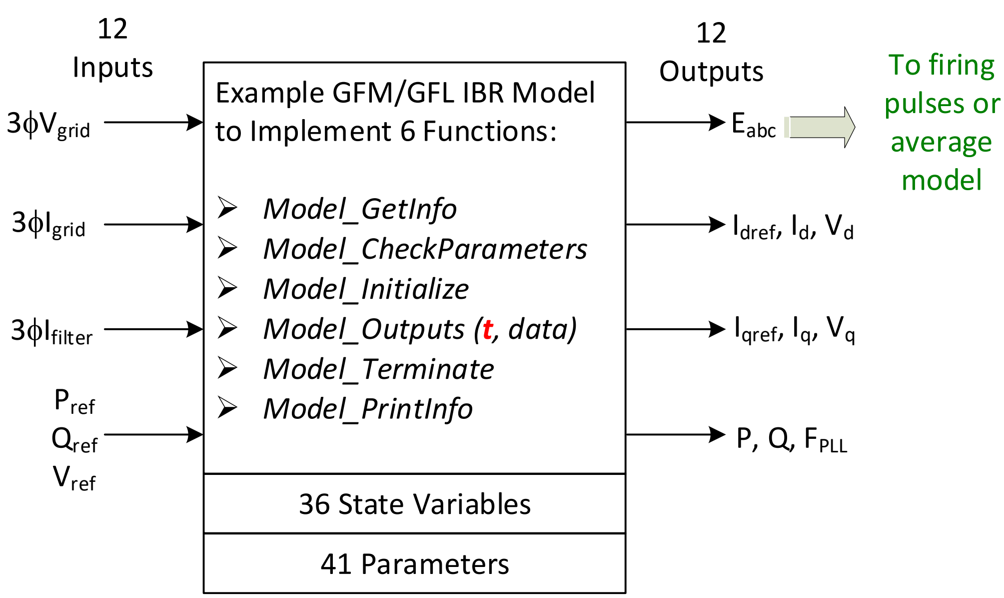


Figure 2.3: DLL Interface for Example IBR Control Code Model of Grid-Forming (GFM) and Grid-Following (GFL) Inverters

[47] The model user must understand the meaning of 12 inputs and 12 outputs in Figure 2.3. These may vary in other examples that meet the DLL specification. The IEEE P3597 might standardize these signal inputs and outputs.

[48] For example, high-frequency harmonics and fast-interaction converter-driven stability are two use cases for detailed switching models in NERC White Paper: Electromagnetic Transient Analysis in Operations Planning, June 2025, https://www.nerc.com/globalassets/our-work/reports/white-papers/whitepaper_-emt-analysis-in-operations_v2.0_final.pdf

Chapter 3: CIM Network Model Support

This chapter describes how the CIM represents lines, transformers, and generator dynamics. These are some of the most familiar network model components. General background on the CIM is detailed as follows:

- The CIM Primer[49] is the best starting point, with a balance between the power system and information system viewpoints. It introduces the Unified Modeling Language (UML), which is the system of diagrams used to specify and document the CIM.
- The Department of Energy (DOE)-funded Data Integration Guide[50] and Application Developer's Guide[51] provide specific information that may be useful in planning a project.
- The CIM Modeling Guide[52] provides essential detail when it comes time to work with the CIM UML.

Lines

EMT supports frequency-dependent and non-transposed[53] line models, which require data on the conductors, bundling, tower cross section, and midspan sags for overhead lines, or geometric and materials information for underground lines. However, frequency-dependent models are not needed for IBR studies, as the frequency range of interest is rather low compared to other EMT applications. Non-transposed line models are not needed for IBR studies because the dynamics are primarily in the sequence components. It is necessary to represent single-line-to-ground faults (SLGF) accurately, but this can be done without using non-transposed line models. For EMT studies of IBRs, it is sufficient to use transposed line models with 60-Hz resistance, inductance, and capacitance values for positive (equal to negative) and zero sequence.

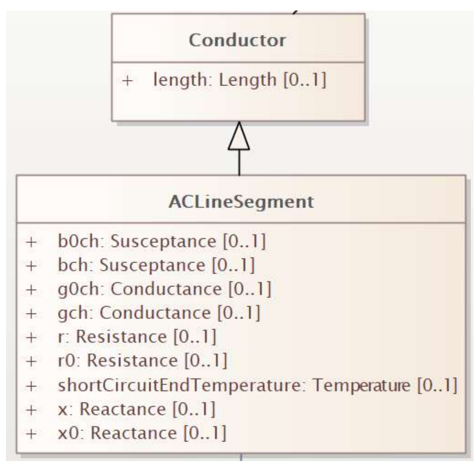


Figure 3.1: CIM Classes for Transposed Lines and Cables

[49] G. Gray, D. Lowe, R. Rhodes, and J. Simmins, "Common Information Model Primer: Ninth Edition," Oct. 2023. Accessed: July 08, 2025. [Online]. Available: https://www.epri.com/research/programs/062333/results/3002026852

[50] A. A. Anderson, E. G. Stephan, and T. E. McDermott, "Enabling Data Exchange and Data Integration with the Common Information Model: An Introduction for Power Systems Engineers and Application Developers," Pacific Northwest National Laboratory (PNNL), Richland, WA (United States), PNNL-32679, Mar. 2022. doi: 10.2172/1922947.

[51] A. A. Anderson, T. E. McDermott, and E. G. Stephan, "A Power Application Developer's Guide to the Common Information Model: An Introduction for Power Systems Engineers and Application Developers – CIM17v40," Pacific Northwest National Laboratory (PNNL), Richland, WA (United States), PNNL-34946, Mar. 2023. doi: 10.2172/2007843.

[52] UCA International Users Group, "CIM Modeling Guide." Accessed: July 08, 2025. [Online]. Available: https://cim-mg.ucaiug.io/latest/

[53] A transposed (or balanced) line model has Z1=Z2, both not equal to Z0. A non-transposed (or unbalanced) line model has coupling impedances between sequence networks. Unbalanced excitations may be applied to both transposed and non-transposed line models. EMT tools generally represent non-transposed lines with phase impedance matrices, or a customized modal decomposition.

The CIM *ACLineSegment* class has positive sequence parameters *r*, *x*, and *bch* in SI units, zero-sequence parameters *r0*, *x0*, and *bch0* in SI units, and a parent class *Conductor.length* attribute in meters.[54] Per-unit impedances from PF or PSPDT models must be converted to SI units for the CIM. Both line charging parameters, *bch* and *bch0*, must be specified as non-zero. The zero-sequence data may be available from SCKT network models. If not, it can be estimated for transmission lines:

$$L = x/(length * \omega) \text{ [H]} \tag{1}$$

Where ω is 377 for 60-Hz systems. If *bch* > 0, estimate the capacitance and surge impedance:

$$C = bch/(length * \omega) \text{ [F]} \tag{2}$$
$$Z = \sqrt{L/C} \text{ [}\Omega\text{]} \tag{3}$$

Otherwise, assuming a typical overhead line surge impedance, estimate *bch* with:

$$Z = 400 \text{ [}\Omega\text{]} \tag{4}$$
$$C = L/Z^2 \text{ [F]} \tag{5}$$
$$bch = C * length * \omega \text{ [S]} \tag{6}$$

Verify that the positive sequence (line mode) propagation velocity does not exceed the speed of light:

$$v = 1/\sqrt{LC} \leq 3e8 \text{ [m/s]} \tag{7}$$

If *v* is 50% or less of 3e8, then the line is probably underground cable, not overhead. If *Z<100* in (3), assume the line is underground cable and estimate zero-sequence parameters by:

$$r0 = r \text{ [}\Omega\text{]} \tag{8}$$
$$x0 = x \text{ [}\Omega\text{]} \tag{9}$$
$$b0ch = bch \text{ [S]} \tag{10}$$

Otherwise, assume overhead line and estimate zero-sequence parameters as follows, using either the PF/PSPDT value for *bch*, or the value estimated from (6):

$$r0 = 2r \text{ [}\Omega\text{]} \tag{11}$$
$$x0 = 3x \text{ [}\Omega\text{]} \tag{12}$$
$$b0ch = 0.6 * bch \text{ [S]} \tag{13}$$

If the line is known to have bundled sub-conductors, then *Z=300* Ω may be used in (4). If the branch consists of two or more lines in parallel, the surge impedance, *Z*, should be reduced accordingly. If the original PF/PSPDT model included line charging data for the branch, then (7) should still help distinguish between overhead and underground lines. If the original data did not include line charging data, then it is probably an overhead line. Cable line charging values are higher and more important than for overhead lines, hence more likely to have been maintained in PF/PSPDT models.

[54] The arrow in Figure 3.1 says *ACLineSegment* "inherits" from *Conductor*. There is also a *DCLineSegment* class in CIM, but it **does not** inherit from *Conductor*. Instead, it has its own *DCLineSegment.length* attribute. This is an example of inconsistency in the CIM that might have arisen from UML development at different times and by different stakeholders. Such inconsistencies are difficult to fix in the full CIM without introducing a "breaking change" for existing implementations. Chapter 4: proposes a mitigation in a simplified CIM profile for EMT IBR models.

Most of the EMT line models selected for IBR studies should include traveling wave behaviors. Traveling wave models are usually more efficient for EMT than a cascade of lumped pi sections, which create internal “buses” and increase the size of the network for numerical solutions. Furthermore, traveling wave models mitigate numerical oscillations that may be triggered from EMT switch operations. The tool may refer to these traveling wave models as Bergeron, Clarke, Karrenbauer, or distributed transposed models. [55] However, if the line is so short that its travel time is less than the simulation time step, a single pi section may be used to represent it. For IBR studies, the simulation time step should not be reduced solely to enable traveling wave model for short line sections. The CIM-to-EMT exporter could make this choice based on a user-supplied Δt.

[55] Frequency-dependent line models require conductor and geometry data not commonly available to large planning models, and relevant only to limited IBR plant issues like harmonics and insulation coordination. This white paper suggests a constant-parameter distributed line model that can be populated from planning model data.

The raw file format does not include a *length* parameter. This calls for a modified procedure to estimate the length from heuristics. The following raw file fragment shows an example transmission line between two buses:

```
0,    100.00, 33, 0, 0, 60.00       / October 01, 2013 18:46:48
 08/25/93 UW ARCHIVE           100.0  1961 W IEEE 118 Bus Test Case
0 / END OF SYSTEM-WIDE DATA, BEGIN BUS DATA
    1,'Riversde    ', 138.0000,2,   1,   1,   1,0.95500,  10.9828
    2,'Pokagon     ', 138.0000,1,   1,   1,   1,0.97139,  11.5228
    #, Name,           kVbase, Area, Zone, ?, ?, Vmag, Vang

0 / END OF GENERATOR DATA, BEGIN BRANCH DATA
    1,     2,'1 ', 0.03030, 0.09990,0.02540,
    From, To, Circuit #, Rpu, Xpu, Bpu
```

The red highlights indicate the system MVA base of 100 and some column fields of interest. Begin with calculating base impedance for bus numbers 1 and 2, followed by the total-length line parameters.

$$Z_{base} = \frac{kV_{base}^2}{MVA_{base}} = \frac{138^2}{100} = 190.44\,\Omega \tag{14}$$

$$R_1 = R_{pu}Z_{base} = 0.0303 * 190.44 = 5.7703\,\Omega \tag{15}$$

$$X_1 = X_{pu}Z_{base} = 0.03990 * 190.44 = 18.8536\,\Omega \tag{16}$$

$$L_1 = X_1/377 = 18.8536/377 = 50.01\,mH \tag{17}$$

$$C_1 = B_{pu}/377Z_{base} = 0.0254/(377 * 190.44) = 0.3538\ \mu\text{F} \tag{18}$$

Then calculate the line surge impedance and total travel time.

$$Z_1 = \sqrt{L_1/C_1} = 376\,\Omega \tag{19}$$

$$\tau_1 = \sqrt{L_1 C_1} = 133\,\mu s \tag{20}$$

From the surge impedance, infer the line is overhead. 138-kV line conductors are not bundled, so estimate the length from a heuristic. (For bundled sub-conductors, use 0.6 Ω/mile instead of 0.8 Ω/mile.)

$$len \approx X_1/0.8 = 23.57 \text{ miles} \tag{21}$$

$$v_1 = len/\tau_1 = 177{,}195 \text{ miles/s} \tag{22}$$

The wave velocity is about 95% the speed of light, which is a plausible and acceptable value. Finally, estimate the zero sequence parameters for EMT. Equations (24)-(25) embed the heuristics from (12)-(13).

$$R_0 = 2R_1 = 11.54\,\Omega \tag{23}$$

$$Z_0 = \sqrt{3/0.6}\,Z_1 = 841\,\Omega \tag{24}$$

$$\tau_0 = \sqrt{3 * 0.6}\,\tau_1 = 178.4\,\mu s \tag{25}$$

For EMT, the total R1 and R0 parameters are converted to per-mile, corresponding to the length estimated in (21).

For the CIM, the total zero sequence inductance and susceptance are obtained from the heuristics, (12)-(13). Then, convert the results to per-unit-length for the CIM.

Transformers

Figure 3.2 shows important considerations for EMT transformer modeling. Legacy PF models may still represent three-winding transformers with separate two-winding transformers in a star equivalent (**Figure 3.2** left) because legacy PF tools only supported two-winding transformers. One of the legacy two-winding transformers is likely to have negative reactance, which is acceptable for PF simulations but not for EMT.[56] Furthermore, the star point is retained in legacy PF models with a base voltage of, typically, 1 kV or 0.1 kV. In preparing such models for EMT, the two-winding transformers must be reaggregated into correct three-winding models.

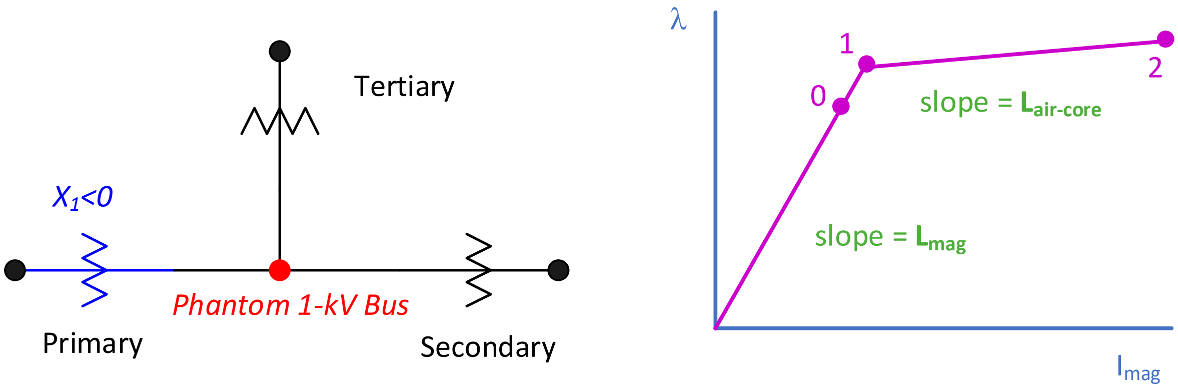
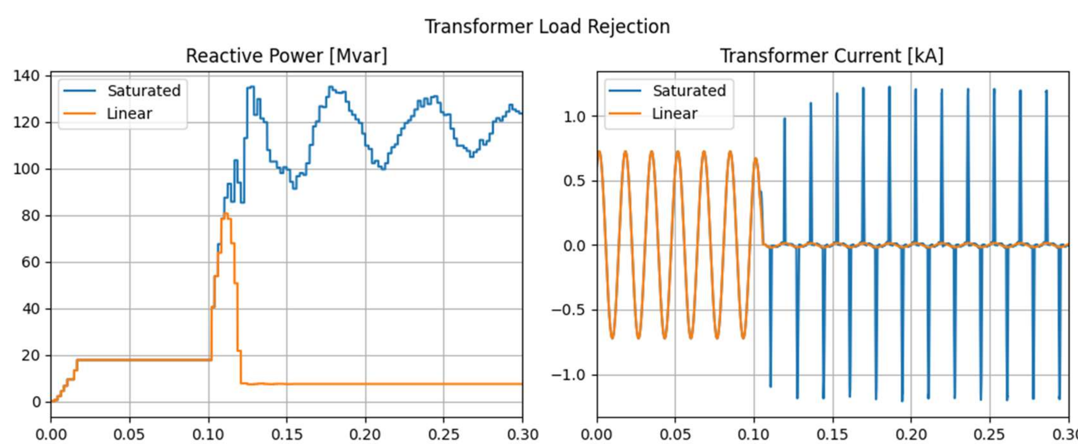


Figure 3.2: Three-Winding Transformer Star Equivalent (Left) and Saturation Characteristic (Right) for EMT [29]

To represent the transformer zero-sequence and negative-sequence behavior for EMT, it is essential to correctly specify the delta or wye connection type for each winding. It is also important to correctly specify any neutral impedance, or ungrounded connection, for wye windings. This data is typically not included with PF or PSPDT models. However, it is always included in SCKT models.

Transformer saturation can also be important to model in EMT studies of IBR because under dynamics conditions, saturation may increase the reactive power[57] demands on generators. **Figure 3.3** shows a severe example of transformer load rejection at the end of a long 345 kV line. Without saturation, Q drops to nearly zero. With saturation, the magnetizing current is high due to overvoltage, and the measured Q is high. Note that Q is ill-defined under these non-linear conditions, but there is still an effect on machines and IBR. Also, the P/Q meter takes a couple of cycles to "wake up" at the beginning of the simulation.



[56] Negative resistance is not allowed in impedance network reduction for EMT. Negative reactance branches may solve, but the EMT tool may not accept such inputs.

[57] It is not rigorous to view transformer saturation current as reactive power, but the practice is widespread.

Figure 3.3: Transformer Load Rejection, With and Without Saturation

For IBR studies, saturation can be modeled with sufficient accuracy by a two-slope characteristic (**Figure 3.2** right). It is possible to use more points and slopes, but the data is not readily available. [58]

Figure 3.4 shows the CIM classes that represent balanced, three-phase power transformers with any number of windings. The CIM also supports modeling from transformer test reports, and unbalanced banks of transformers, (e.g., from a single-phase replacement), but these are not considered here. The *TransformerMeshImpedance* represents the short-circuit impedance between pairs of windings, converted from per-unit to ohms with reference to one of the windings. The *TransformerCoreAdmittance* attributes represent magnetizing and core loss current, converted to ohms and mhos with reference to one of the windings. Each winding corresponds to a *PowerTransformerEnd* in the CIM, with attributes for rated voltage, *ratedU*, rated apparent power, *ratedS*, and the *WindingConnection*. The windings also inherit grounding attributes from *TransformerEnd*. Each winding has an *endNumber*, from 1 to the number of windings. The windings should be numbered in order of decreasing *ratedU*.

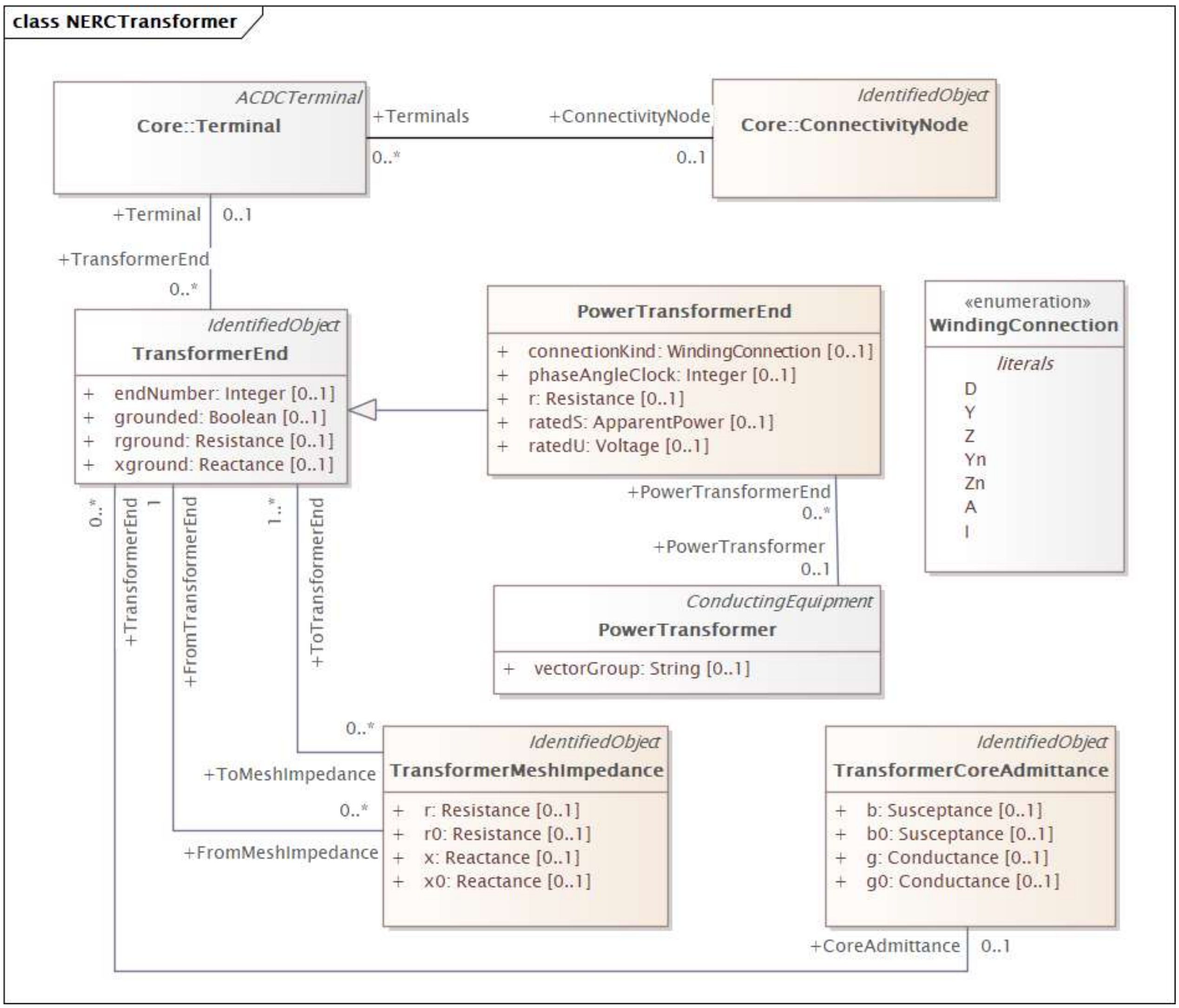


Figure 3.4: CIM Classes for Balanced Three-Phase Transformers

[58] Remnant core flux should be considered in studies of black-start, inrush, and DC offset. Remnant flux is an EMT initial condition that does not affect CIM network modeling, nor is it determined by power flow initial conditions.

Figure 3.4 also shows the CIM *Terminal* class. There is one *Terminal* associated to each *TransformerEnd* (winding), two *Terminal*s associated to each *ACLineSegment*, and one to each load, shunt or generator, etc. The *Terminal*s then associate to a *ConnectivityNode*, which is the "bus" in the CIM. In other words, branches in the CIM do not connect directly to buses. In legacy power system file formats, it is typical to have the bus number(s) included with the branch or shunt records. One reason for the separate *Terminal* class in the CIM is to provide a distinct connection point for measurements. It may not be necessary for the EMT/IBR use case.

EMT saturation models are generally defined in characteristics of instantaneous flux linkage, λ, vs. instantaneous magnetizing current, I_{mag}. The CIM *TransformerCoreAdmittance* class has a *b* attribute for magnetizing susceptance and a *g* attribute for core losses, both in SI units. If these are not available from PF/PSPDT data, then typical non-zero values should be assumed, (e.g., 1% magnetizing current and 0.25% core loss current) before converting those to SI values. Use the secondary, or lowest-voltage winding, excluding the tertiary, for connecting the saturation branch. The short-circuit inductive reactance, *x*, between primary and secondary windings is also needed, with *x* in ohms referred to the secondary winding. The secondary winding's rated voltage, *u*, should be expressed in peak volts, not RMS volts. The steady-state magnetizing current, I_0, should be expressed in peak amps, not RMS amps, referred to the secondary winding. From this information, we can estimate the slopes and point **0** in Figure 3.2 (right) as follows:

$$L_{mag} = b/\omega \text{ [H]} \tag{26}$$
$$L_{air-core} = 2x/\omega \text{ [H]} \tag{27}$$
$$\lambda_0 = u/\omega \text{ [V-s]} \tag{28}$$

Then we choose a typical "knee voltage," V_{knee_pu}, for the transformers, typically 1.05 to 1.15 per-unit. This defines a straight-line extrapolation from point 0 to point 1 in Figure 3.2 (right). Then determine points **1** and **2** as follows:

$$I_1 = V_{knee\_pu} * I_0 \text{ [A]} \tag{29}$$
$$\lambda_1 = V_{knee\_pu} * \lambda_0 \text{ [V-s]} \tag{30}$$
$$\Delta I = 1000 \text{ [A]} \tag{31}$$
$$I_2 = I_1 + \Delta I \text{ [A]} \tag{32}$$
$$\lambda_2 = \lambda_1 + L_{air-core}\Delta I \text{ [V-s]} \tag{33}$$

EMT saturation models typically take points **1** and **2** for input, with the origin implied. The steady-state solution may use point **0**. In the linear range of operation, this will be consistent with points **1** and **2** after time-stepping begins.

Dynamics

Rotating machine dynamic models in EMT and the CIM are usually identical to the *dq0* dynamic models that are common to PSPDT tools. Control systems for machines, such as the excitation systems, automatic voltage regulators, governors, and power system stabilizers are represented as block diagram models in PSPDT tools and are represented identically in EMT software.[59] There are some cases where additional model parameters would be useful to model in EMT tools that are not supported in PSPDT tools, such as additional points on the saturation curve, but such extensions are out of the scope of this effort. No further extensions are needed in the CIM to support these conventional block diagrams or machine models. Some EMT tools may not implement all models found in a PSPDT tool's library, but this would not be due to a mismatch in data.

For inverter-based models, the differences between the PSPDT model and the EMT model can differ substantially, with EMT tools being capable of modeling the power electronic circuitry, DC bus, and control loops in full detail. This model representation is commonly referred to as a "switching model." Extensions are needed to fully support switching models, as described in the next section. However, the existing PSPDT models, commonly referred to as

[59] In CIM Dynamics and other tool-specific data formats, legacy terms like "governor" are still used, even though modern implementations are typically digital instead of electromechanical. In context, these are all simply controllers and physical dynamic processes to be modeled.

"average models," are supported by the CIM without extensions. The average models are still useful in EMT simulations in some cases, for example in instances where the full IBR switching model is not available or the model is placed electrically distant from the area under study.

The CIM has a *RemoteSignal* mechanism for connecting the dynamic control block diagrams to various points on the network, (i.e., outside of the rotating machine). Many kinds of remote signal are available from an enumerated data type (e.g., voltage magnitude, current magnitude, frequency, frequency deviation). It is necessary to connect the *RemoteSignal* to a *Terminal*. This mechanism could be simplified, as discussed in **Chapter 4:**.

Figure 3.5 shows the current CIM Dynamics support for WECC generic models, including some of the newer DER aggregation models, in the form of a UML schema. In the CIM, the inverter is called a *PowerElectronicsConnection*, which is part of the *Wires* package. It has a *Terminal* for connection to other grid components (e.g., the *Unit Transformers* from **Figure 2.2**). A similar pattern is used for connecting exciter, governor, and power system stabilizer dynamics to a *SynchronousMachine* instead of *PowerElectronicsConnection*. No further extensions are needed to support EMT studies with WECC models, as demonstrated in [29].

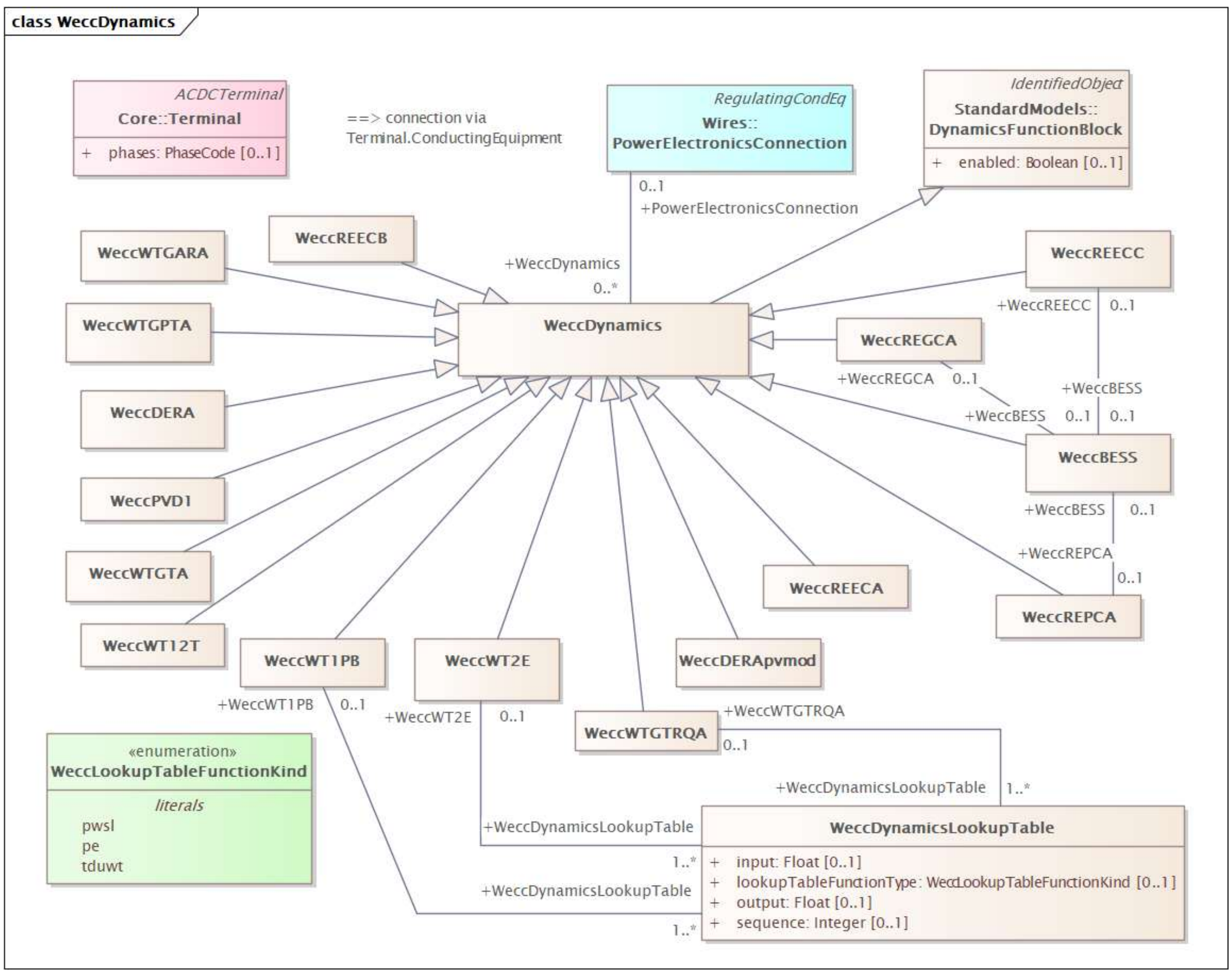


Figure 3.5: CIM Dynamics UML for the WECC Generic IBR Models

Chapter 4: CIM Extensions for IBR EMT Studies

This chapter discusses main areas for extending the CIM to cover the EMT IBR use case.

Extensions for Transformer Saturation

Figure 4.1 shows the UML for a *TransformerSaturationCurve* with attributes defined in **Table 4.1**. The UML uses a well-established pattern in the CIM by associating an optional *TransformerSaturationCurve* to the *TransformerCoreAdmittance*. There will be two instances of *CurveData* for each saturation characteristic, defining points **1** and **2** of **Figure 3.2** (right) with equations (29), (30), (32), and (33). This UML supports any number of points on a saturation characteristic if the data is available.

Table 4.1: CIM Attributes for Transformer Saturation Curve

Name	CIM Type	Description
Curve		
xUnit	UnitSymbol	A (current)
y1Unit	UnitSymbol	Vs (flux linkage, choosing Vs over Wb units)
CurveData		
xvalue	Float	Instantaneous current, Imag
y1value	Float	Instantaneous flux linkage, lambda

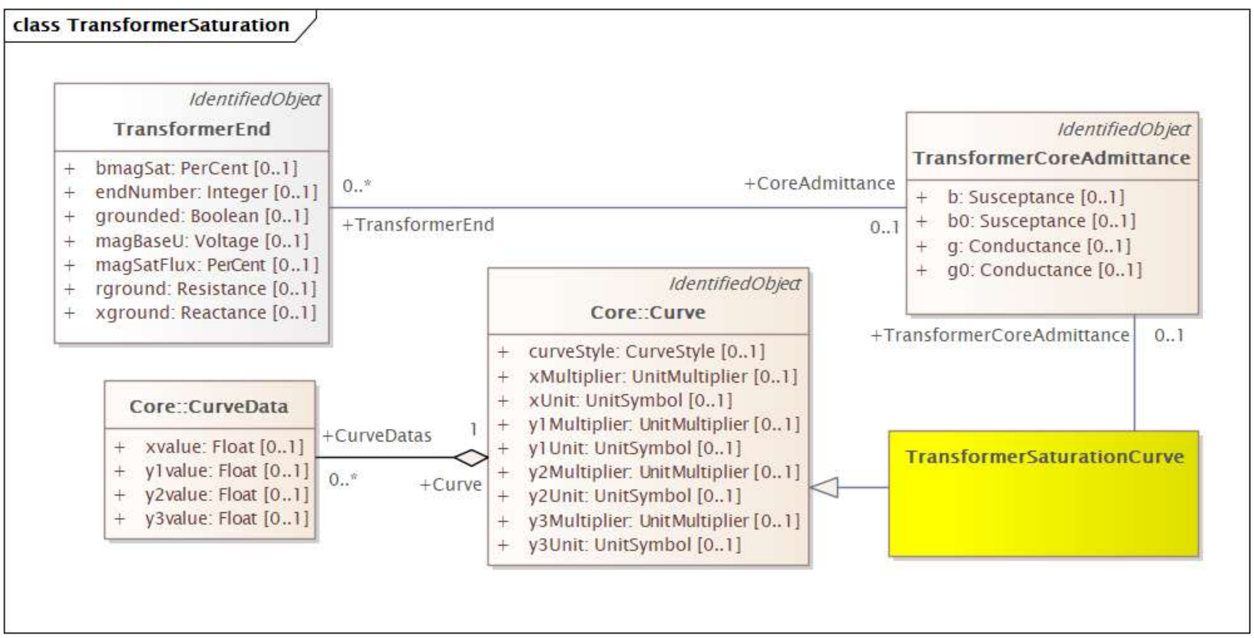


Figure 4.1: CIM Extension UML for Transformer Saturation Added to Wires:TransformerCoreAdmittance, New Class in Yellow

Extensions for Exact IBR Control Code

The IEC CIM working groups have considered "functional modeling" of control system logic in the CIM but have not completed the task. This sub-chapter proposes to leverage the real-code DLL interface specification [14] for both proprietary and possible new generic models [35]. Minimal CIM extensions are proposed to maintain DLL location, version, and parameter sets. Even if developed, a CIM-centric functional modeling feature may be limited in function, require additional learning, and fail to hide proprietary details.[60] The DLL interface has none of these defects.[61] It enables the customization of CIM to handle average or switching converter models, plant-level controls, real hardware control code, and other gaps in present IBR modeling practices. Any kind of model, e.g., large loads, could be implemented through the DLL interface.

Figure 4.2 shows new IBR class associations on a UML diagram, with new attributes as defined in **Table 4.2**, to complete representation of the orange highlights in **Figure 2.2** for the CIM. Two of these, *IEEECigreDLL* and *InverterBasedResourcePlant*, are optional associations to a *PowerElectronicsConnection*, and they inherit from *PowerSystemResource* and *IdentifiedObject*. The *name* and *mRID* attributes of *IdentifiedObject* support unique and persistent identifiers for the new classes.

The *IEEECigreDLL* attributes in **Table 4.2** can specify the location and vintage/version of a DLL, which is assumed to comply with a standard interface specification for the DLL. There should be no attempt to model details of the DLL interface in the CIM. Instead, the specification [14] should be incorporated by reference. Model builders and EMT tool vendors would be responsible for using the specification to connect the DLL to the inverter bridge, and other components, at simulation time.[62] The class attributes include the model's expected *timeStep* and its support for EMT and/or root mean square (RMS) simulation modes. Note that RMS is the same as PSPDT in this context for the DLL. These *timeStep*, *supportsEMT*, and *supportsRMS* attributes are discoverable from the model's application program interface (API) at runtime, so they are not strictly necessary in the UML. However, they are useful attributes for the repository to determine whether a DLL is compatible with an EMT network simulation before launching it.

Table 4.2 includes a starting point for attributes of *InverterBasedResourcePlant*, primarily focused on the passive filter components in **Figure 2.2**. This list would be refined in collaboration with EMT tool vendors and other interested parties. For example, *switchingFrequency=0* implies use of an average source model, but the CIM working groups generally prefer a more explicit Boolean attribute for such choices. It is expected that a tool-neutral set of attributes can be developed for *InverterBasedResourcePlant*, based on experience with other use cases (e.g., static var compensation (SVC) and HVdc transmission, which were successfully represented in the CIM).

Table 4.2: CIM Attributes for two new IBR Plant Classes

Name	CIM Type	Description
IEEECigreDLL		
vendorName	String	Name of the model provider
firmwareVersion	String	Version of the hardware provider's firmware
firmwareInstalled	DateTime	Date and time of the firmware installation at the IBR plant

[60] For example, many vendors and researchers have already learned and adopted the MATLAB/Simulink family, Modelica, or another package with comparable functionality. MATLAB/Simulink models, with control logic, can be compiled to C code for the DLL interface. Similarly, tool-specific user code models might be migrated to compiled code that conforms to the new DLL interface specification. CIM-centric functional modeling is not required for any of these use cases.

[61] Because it comprises executable code, a DLL opens up software vulnerabilities to address in IEEE P3743.

[62] CIM for Dynamics already represents *RemoteSignals* as needed for *SynchronousMachine* control. Extensions may (or may not) be required for IBR plants, but the details are to be determined.

Table 4.2: CIM Attributes for two new IBR Plant Classes		
Name	**CIM Type**	**Description**
Uri	String	Location of the DLL, either a universal resource locator or network-accessible file name.
timestep	Float	Hard-wired simulation time step for this DLL
supportsEMT	Boolean	True if the DLL supports EMT simulation
supportsRMS	Boolean	True if the DLL supports RMS simulation
InverterBasedResourcePlant		
dcLinkVoltage	Voltage	Voltage of the direct current (dc) bus, if applicable
dcLinkCapacitance	Capacitance	Capacitance (plus-to-minus) of the dc bus, if applicable
acFilterRgrid	Resistance	Grid-side resistance of the alternating current (ac) tee filter
acFilterLgrid	Inductance	Grid-side inductance of the ac tee filter
acFilterCapacitance	Capacitance	Shunt capacitance of the ac tee filter
acFilterRbridge	Resistance	Inverter bridge-side resistance of the ac tee filter
acFilterLbridge	Inductance	Inverter bridge-side inductance of the ac tee filter
switchingFrequency	Frequency	PWM switching frequency

One of the new classes in **Figure 4.2**, *IEEECigreDLLValue*, supports the DLL parameter sets with saved values. The values may be 1 of the 10 enumerated *IEEECigreDLLParameterKind* selections, stored in the *value* attribute in string format. The *sequenceNumber* of each value must match the expected order in the model's API. The *sequenceNumber* and *parameterType* can be determined from the API at runtime, but including these in the UML allows the parameter values to be saved in the EMT model repository. The API also provides access to parameter names, default values, minimum values, and maximum values, but these attributes are not essential to maintain the repository, so they are left out of the proposed UML.[63] To obtain parameters for calling a DLL at runtime, one would query the table of *IEEECigreDLLParameter.* A Structure Query Language (SQL) "WHERE" clause would be used to match the *PowerElectronicsConnection* as a foreign key. An SQL "ORDER BY" clause would be used to sort the return values by *sequenceNumber*. Processing each row of the query result, one would check the *parameterType* attribute to properly convert the string *value* into a 32-bit integer, 64-bit floating point number, etc.

[63] An alternative design would store each parameter value in 1 of 10 concrete classes. There would be one concrete class for each *IEEECigreDLLParameterKind* enumeration. These concrete classes would inherit from *IEEECigreDLLValue*. Each one would have a specific typed *value* attribute that matches the kind of DLL parameter. In this alternative design, there would be no *parameterType* attribute of *IEEECigreDLLValue*. Furthermore, the *IEEECigreParameterKind* enumeration becomes unnecessary. This alternative has some advantages, especially for object-oriented databases and SPARQL queries. With a relational database and SQL queries, this alternative is more complicated because ten different tables would have to be queried to obtain all the DLL parameters. The latest version of CIM UML shows another alternative to consider, in the *DetailedModelDescription* package of *Dynamics*.

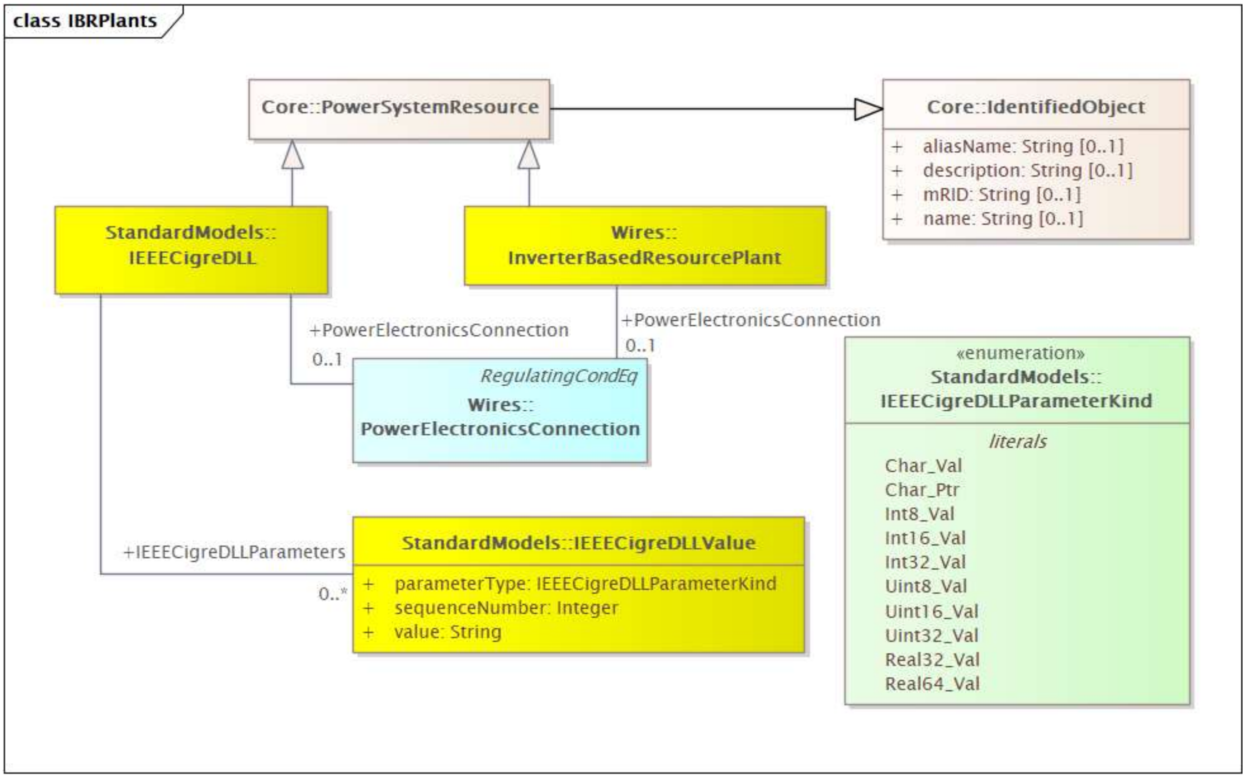


Figure 4.2: CIM Extension UML for Plant Parameters and DLLs Added to Wires:PowerElectronicsConnection. New Classes are Shown in Yellow, and a New Enumeration is Shown in Green

The DLL API includes *ExternalInputs* and *ExternalOutputs* that must be connected to other parts of the network model. Some of that information could be represented in **Figure 4.2**, but that would not suffice for connecting these interface variables to other parts of the network model. Some of these inputs and outputs may be voltages and currents at network terminals, but others are not. It is up to the simulator, probably with user intervention, to make the right connections. For some use cases, the inputs and outputs could be further standardized, (e.g., as in **Figure 2.3**). Future extensions of the UML may be developed to support this task.

Review of CIM Classes for EMT

Table 4.3 shows the CIM class counts in a WECC 240-bus model with IBR added [31]. There are 243 instances of the *ConnectivityNode* class, which roughly corresponds to the number of buses, nominally 240. There are 320 instances of *ACLineSegment*, corresponding to overhead lines and cables. This example has 139 loads in the *EnergyConsumer* class, 111 synchronous machines, 25 WECC generic photovoltaic (PV) plants, and 10 WECC generic wind plants. There are 9 series capacitors and 8 shunt reactors. There are 122 instances of *PowerTransformer*, corresponding to transformers, all of them two-winding in this example. The same type of governor was used for hydro and steam units, which is considered an error but not necessary to correct for this discussion. The remainder of this sub-chapter discusses opportunities for simplification of the CIM classes appearing in **Table 4.3**.

Table 4.3: CIM Class Counts in a WECC 240-Bus Model

CIM Class	Count	Description
ACLineSegment	320	Overhead lines and cables
BaseVoltage	7	System base voltages that are referenced

Table 4.3: CIM Class Counts in a WECC 240-Bus Model		
CIM Class	**Count**	**Description**
ConnectivityNode	243	Buses
CoordinateSystem	1	Reference system for X-Y positions
EnergyConsumer	139	Loads
EquipmentContainer	1	The BPS
ExcST1A	111	Exciters for hydro, nuclear, and thermal generators
GovSteamSGO	111	Governors for hydro, nuclear, and thermal plants
HydroGeneratingUnit	16	Prime movers for hydro generating plants
LinearShuntCompensator	7	Shunt capacitor or SVC
LoadResponseCharacteristic	1	A set of constant impedance, constant current, constant power (ZIP) load coefficients
Location	743	A set of X-Y locations for a significant component, e.g., series or shunt. Contains one or more PositionPoints
NuclearGeneratingUnit	2	Prime movers for nuclear plants
PhotovoltaicUnit	25	Power sources for solar photovoltaic plants
PositionPoint	1194	X-Y locations, one per terminal in this example
PowerElectronicsConnection	35	Inverters for solar and (Type 4) wind plants
PowerTransformer	122	Transformer units or banks, associated with windings, core admittances, and mesh impedances
PowerTransformerEnd	244	Transformer windings, at least two per transformer
PssIEEE1A	111	Stabilizers for hydro, nuclear, and thermal generators
SeriesCompensator	9	Series capacitors
SynchronousMachine	111	Rotating machines, e.g., thermal, hydro, nuclear
Terminal	1194	One for each branch end, including shunts, plus transformer windings
ThermalGeneratingUnit	93	Prime movers for thermal plants, i.e., fossil-fueled
TransformerCoreAdmittance	122	Represents magnetizing current and core losses, one per transformer
TransformerMeshImpedance	122	Short-circuit impedance between a pair of transformer windings
WeccREECA	35	WECC generic IBR plant PQ and current limit controllers

Table 4.3: CIM Class Counts in a WECC 240-Bus Model		
CIM Class	**Count**	**Description**
WeccREGCA	35	WECC generic IBR converters
WeccREPCA	35	WECC generic IBR plant controllers
WeccWTGARA	10	WECC generic wind plant aerodynamics
WeccWTGTA	10	WECC generic wind plant drive trains
WindGeneratingUnit	10	Power sources for wind plants

<u>Simplifying database queries and results</u>. The CIM is an object-oriented design, making use of concepts like inheritance and polymorphism. For example, in **Figure 3.5**, there are 15 classes that inherit from *WeccDynamics*, indicated by lines with open arrowheads. There is an association (line with no arrows) from *WeccDynamics* to a zero-or-one *PowerElectronicsConnection* (i.e., an inverter). For ease of use, the association is typically used, (i.e., profiled) in the direction from the *0..** end to the *0..1* end. Traversing that association in the opposite direction is zero-to-many and polymorphic. In other words, an association from *PowerElectronicsConverter* to *WeccDynamics* could be to any of those 15 descendant classes, to be determined by parsing the database query results. These concepts of inheritance and polymorphism do not translate well to the relational database schemas that have been used for decades along with SQL. Many implementations of the CIM use a NoSQL database technology, such as a "triple store" or "graph database." The query language for those technologies may be SPARQL instead of SQL, which creates a learning curve on top of a different database management system to install. This report proposes to avoid NoSQL databases by avoiding inheritance and polymorphism (i.e., these concepts are still in the CIM but will not appear in the EMT-IBR profile).

Nearly all associations in the CIM UML can be represented with foreign keys in a relational database. An exception occurs with many-to-many associations, which generally require a new "join table" to implement in SQL. Many-to-many associations are also more complicated in NoSQL, at least in terms of parsing the query results. But many-to-many associations will be avoided in the proposed EMT-IBR profile. They are already minimized in the CIM. As a result, the proposed schema can be published in SQL's Data Definition Language (DDL), and other formats like Extensible Markup Language (XML) Schema Definition (XSD) and JavaScript Object Notation (JSON) Schema.

The EMT-IBR profile would always be derived from the full CIM UML. This would simplify the upgrade path to full CIM, for users of the EMT-IBR profile who later wish to use the full CIM. This might occur for those who later adopt more CIM use cases than just EMT studies of IBR.

<u>Flattening the *ACLineSegment* class</u>. **Figure 3.1** showed *ACLineSegment* inheriting from *Conductor*, although *DCLineSegment* does not inherit from *Conductor*. Both line segments inherit from *IdentifiedObject*, too, as do most classes in the CIM UML. To remove the concept of inheritance from an EMT-IBR profile, the attribute names must be "flattened." CIM profiles typically qualify all attribute names with the name of the CIM class where the attribute was defined, somewhere up the chain of inheritance. This is not strictly necessary, by Rule 60 of the CIM modeling guide, which seems to prohibit conflicting attribute names [40]. **Table 4.4** illustrates the approach. Use of the flattened attribute names will shield adopters from inconsistency in the *length* attribute and from the inheritance of *IdentifiedObject* along with many other inherited classes. If any attribute names are found to conflict, the CIM would be extended and a "fix" proposed to the CIM User's Group.

Table 4.4: Flattened Attribute Names for ACLineSegment and DCLineSegment

Fully qualified Name	Flattened Name
ACLineSegment	
IdentifiedObject.name	name
IdentifiedObject.mRID	mRID
Conductor.length	length
ACLineSegment.r	r
ACLineSegment.x	x
ACLineSegment.bch	bch
ACLineSegment.r0	r0
ACLineSegment.x0	x0
ACLineSegment.bch0	bch0
DCLineSegment	
IdentifiedObject.name	name
IdentifiedObject.mRID	mRID
DCLineSegment.length	length
DCLineSegment.resistance	resistance
DCLineSegment.inductance	inductance
DCLineSegment.capacitance	capacitance

Names and identifiers. The inherited *IdentifiedObject* class defines two attributes proposed for use in the EMT-IBR profile. The *name* is human-readable and not necessarily unique. The *mRID* is a “master resource identifier” that must be unique among all objects in the network model. Bus numbers in a typical “raw file” could meet the uniqueness requirement because there are no branch numbers, generator numbers, or load numbers. Each of those series and shunt components is defined in the raw file by one or two bus numbers, possibly supplemented with a parallel branch number. According to the CIM, however, those other components need their own unique identifiers, and the bus numbers shall not be re-used for them. Approaches to satisfy this requirement include:

1. Construct a composite *mRID* for series and shunt components, constructed from bus numbers and parallel branch numbers according to some defined rule. This approach seems brittle and may require extra parsing for practical use of the constructed *mRID*s.
2. Don’t include *mRID* in the profile for series and shunt components, only for the *ConnectivityNode* (i.e., bus). This forgoes the important benefits of unique identifiers for every model component. It is a feasible barebones approach if the next option is not adopted.

3. Follow the recommended full-CIM practice of using universally unique identifiers (UUID)[64]. The UUID is a 128-bit value in a defined format (e.g., AC8407E8-8E9C-43F7-8617-DA906382C956). Each UUID is generated once by an algorithm then maintained by a "model authority" according to the CIM. This practice ensures that models can be stitched together without worry that both model creators used "bus 100" for different physical buses. It ensures that buses and components retain their unique identities throughout the life of a network model, which is typically decades. Within the four major North American Interconnections (Eastern, Western, Texas, Québec), this "model authority" function is already met, in part, by the need to maintain unique bus numbers within the interconnection. To satisfy the CIM requirement, unique identifiers would be maintained <u>between</u> the interconnections, also for series and shunt components. In Europe, this has already been done for about 30 different grid operators.

<u>The need for *PowerTransformerEnd* class</u>. **Figure 3.4** shows the CIM UML for a *PowerTransformer* comprising two or more windings, or *PowerTransformerEnd*s in the CIM. Each winding has its own attributes for voltage rating, apparent power rating, winding connection type, grounding, and connection to a *Terminal*. The short-circuit impedances are defined by *TransformerMeshImpedance* connected between pairs of windings, i.e., $n(n-1)/2$ instances where n is the number of windings.

The associations are defined toward the ends with cardinality of 1 or 0..1. Each *PowerTransformerEnd* associates with the *PowerTransformer* to which it belongs and to the *Terminal* to which it is connected. Each *TransformerMeshImpedance* associates with two *PowerTransformerEnd*s.[65] Those associations are all simple foreign keys in SQL. There will be four separate transformer tables in the relational database, used as follows:

1. Query the ***PowerTransformer*** table for a list of all transformers. Process the query result one row at a time. The *vectorGroup* attribute should indicate the number of windings.
2. For each *PowerTransformer* row, query the ***PowerTransformerEnd*** table for all windings having that *PowerTransformer* as a foreign key. This confirms the number of windings. For each row in the query result:
   a. The flattened attributes from *TransformerEnd*, *PowerTransformerEnd*, and *IdentifiedObject* will be available.
   b. There should be a foreign key into the *Terminal* table, as described in the next subsection, which also proposes the use of *ConnectivityNode* instead of *Terminal*.
   c. Query the ***TransformerCoreAdmittance*** table for exciting branches having this *PowerTransformerEnd* as a foreign key. If there is one, the exciting branch connects to this winding. If saturation is included in the EMT-IBR profile, process **Figure 4.1** for the nonlinear characteristic. Another option is to add four attributes to the *TransformerCoreAdmittance* by extension, which will be simpler for only two points.
   d. Query the ***TransformerMeshImpedance*** table for branches having this *PowerTransformerEnd* as a foreign key, either as *FromTransformerEnd* or *ToTransformerEnd*. This process builds up a set of mesh impedances between pairs of windings.
   e. After completing steps a–d for all windings, there will be enough information to create the EMT model for this *PowerTransformer*.

Rather than issuing many queries individually, it could be more efficient to query each of the four tables one time, storing the results in memory. This is strictly a matter of implementation. Either way, the process is more complicated

[64] P. J. Leach, R. Salz, and M. H. Mealling, "A Universally Unique IDentifier (UUID) URN Namespace," Internet Engineering Task Force, Request for Comments RFC 4122, July 2005. doi: 10.17487/RFC4122.

[65] The *ToTransformerEnd* in **Figure 3.4** has cardinality 1..*, for the case of a short-circuit test between one energized transformer winding, and two (or more) grounded windings. Sometimes, tests like that are done for transformers with two identical secondary windings. Rather than support that use case, it may be proposed that such data be converted to strictly pairwise mesh impedances before storing them in CIM.

than processing transformer data in raw files, one row at a time. The major benefit of CIM transformer modeling is that it works for any number of windings.

The CIM also provides for unbalanced transformer modeling, as typical for distribution system transformers, and for EHV banks that have a different spare on one phase. The CIM also provides for transformer test data that could be shared among transformers in a library. Note that in **Figure 3.4**, the *TransformerMeshImpedance* and *TransformerCoreAdmittance* classes cannot be shared as library elements. These features are not proposed for the EMT-IBR profile but remain available in the full CIM for later adoption.

Removing the *Terminal* and *TopologicalNode* classes. The *Terminal* class appears in **Figure 3.4**, **Figure 3.5**, and many other parts of the CIM UML. There are 1,194 instances of *Terminal* in **Table 4.3**. As shown in **Figure 3.5**, the CIM connects components to a *ConnectivityNode* (bus) through a *Terminal* rather than directly. This typically creates many instances of *Terminal*, each of which has a *name* and unique identifier or *mRID*. This adds complexity to the process of connecting network model components from SQL query results. According to EPRI's "Common Information Model Primer, Eleventh Edition"[66], the main reason for including *Terminal* in CIM was to provide well-defined connection points for measurements like voltage transformer (VT) and current transformer (CT). The original CIM use cases focused on energy management system (EMS) applications, including topology processing, state estimation, and representing real network measurement points. The use case for *Terminal* in EMT modeling is not so strong for the following reasons:

1. The EMT analyst can request simulation outputs from any needed point, not just where physical sensors are connected. Sometimes an EMT study will address CT characteristics, VT characteristics, communication channel performance, signal processing, and protection logic, but none of these details are included in the CIM. The analyst would have to create those sub-models manually. Furthermore, such details are unlikely to be part of IBR planning and impact studies. These factors reduce the need for a CIM *Terminal* class.
2. EMT models are more likely to require node-breaker connections, rather than bus-branch, even though only bus-branch may be available from the PSPDT or PF data. The EMT analyst may need to add switchgear manually in at least part of the network model. In the CIM, there is a polymorphic *Switch* class with about eight descendants, including *DisconnectingCircuitBreaker*. The *Switch* class descendants all have two *Terminal* connections and hence two *ConnectivityNode* connections. In a node-breaker model, the *ConnectivityNode* will have a closer relationship to the *Terminal* and loses nothing with respect to switchgear modeling with *Terminal*s.
3. The CIM *Terminal* class includes an inherited *sequenceNumber* attribute, so there is already an ordering concept for *Terminal*s. In addition, the CIM defines positive branch flow measurements as "into" the branch. That means, for example, that a CT polarity dot is always facing out from the branch. That may not match the polarity of physical CTs connected to a component. This does not matter very much for CTs connected to a zero-impedance (in the CIM) element like the *Switch* descendants. The charging current would matter for CTs on *ACLineSegment*. The main point is that a CIM *Terminal* does not improve model precision for EMT.

The CIM includes a *TopologicalNode* class to support network reduction for state estimation. However, state estimation is usually not done in EMT. As a result, *ConnectivityNode* generally has a one-to-one correspondence with *TopologicalNode*. The *TopologicalNode* is preferred by some CIM practitioners but was already removed from the example in **Table 4.3** for EMT.

As a CIM extension, *bus1* and *bus2* attributes should be added to all two-terminal branches, like *ACLineSegment*. They will be associations (foreign keys) to a *ConnectivityNode*. A similar *bus1* attribute should be added to one-terminal branches like *PowerTransformerEnd*, *EnergyConsumer*, *PowerElectronicsConnection*, *SynchronousMachine*, etc.

[66] EPRI, "Common Information Model Primer: Eleventh Edition," 3002032389, Sept. 2025. Accessed: Oct. 24, 2025. [Online]. Available: https://www.epri.com/research/products/3002032389

Another option is to add the *bus1* and *bus2* attributes to ancestor classes like *ConductingEquipment*, *EnergyConsumer*, and/or *EnergyProducer*. The IEEE P3743 Working Group will discuss this in more detail before adoption, but it is expected that the reduction in profile complexity will be worthwhile.

Removing the *Terminal* and *TopologicalNode* classes will create "breaking changes" in the CIM. That means existing CIM software may not be able to properly read the proposed files. These breaking changes are necessary to support widespread adoption of the simplified approach, which is optimized for EMT IBR.

<u>Possibly removing the polymorphic *SynchronousMachineDynamics*, *WeccDynamics,* and *RemoteInputSignal* classes</u>. These are polymorphic classes that complicate the processing of SQL query results from a relational database, as illustrated in the discussion of *WeccDynamics* in **Figure 3.5**. Removing these polymorphic classes will replace one association with 15 in the case of WeccDynamics, with about 5 in the case of *RemoteInputSignal*, and possibly hundreds in the case of *SynchronousMachineDynamics* (for all the types of exciters, governors, stabilizers in current use). All these tables might have to be queried in building the EMT model, which may still be simpler than polymorphic queries with SQL.

Instead of connecting to a *Terminal*, the *RemoteInputSignal* would connect to a *ConnectivityNode*, as across variables like voltage and frequency, this creates no ambiguity. For through variables like current, active power, and reactive power, it may be necessary to allow *RemoteInputSignal* connections to a component (not bus) having the new *bus1* and *bus2* attributes. The positive sign convention would be from *bus1* to *bus2*, possibly modified by a polarity attribute added to *RemoteInputSignal*.

The IEEE P3743 Working Group should explore options to reduce the overall complexity of querying for dynamics data before making the final decision on removing polymorphic classes.

<u>Diagram layout and the *PositionPoint* class</u>. This class has *x*, *y*, and *z* location attributes for association to a *Terminal*. In the demonstration EMT models to date, these locations represent geographic layouts, as is typical for unbalanced distribution feeders that have geographic information system (GIS) data. There is another CIM feature for one-line diagram layouts, not included in **Table 4.3**. CIM can also represent a sequence of locations for detailed geographic routing of a single component. If the *Terminal* class is removed, the *PositionPoint* class might be associated to a *ConnectivityNode* instead. When EMT tools have imported one-line diagram data from PSPDT files, in some cases, the imported layout may be too cluttered for managing circuit breaker connections and other details of the substations. The CIM won't necessarily solve this problem, especially when the original data source comes from a PF, SCKT, or PSPDT file.[67]

<u>Use the *StateVariables* package for initial conditions</u>. Raw files may include bus voltage and magnitude with bus data, and generator active and reactive power injections with generator data, both from power flow solutions. The CIM has a more flexible option in the *StateVariables* package. It is proposed to use that package, rather than add attributes for initial conditions to other CIM classes like *ConnectivityNode*.

<u>Use the *OperationalLimits* package for branch ratings</u>. Raw files may include attributes for line and transformer ratings (e.g., normal, long-term overload, short-term overload). The CIM has a more flexible option in the *OperationalLimits* package. It is proposed to use that package rather than add attributes for overload limits to other CIM classes like *ACLineSegment*.

[67] Two sample BPS models have been laid out using the Kamada-Kawai method available in Python's *networkx* package. The layouts are weighted by estimated physical line lengths of the branches and their topological connections, but no *xy* coordinates were provided to the algorithm. The resulting WECC 240-bus layout is interesting. The "donut hole" naturally appeared in the middle of the autogenerated layout. Wind plants from the Pacific Northwest appeared at the northern edge rather than the northwest corner, so the layout algorithm rotated the entire network about 45 degrees clockwise. See https://github.com/GRIDAPPSD/CIMHub/tree/feature/SETO/BES.

Identifying IBR plant components. The IBR balance-of-plant components, indicated with green shading in **Figure 2.2**, should be identifiable as a grouping in the EMT-IBR profile. One simple way of doing this is to define a table in SQL DDL with three columns:

1. The *mRID* of a *PowerElectronicsConnection*, which identifies the voltage source converter (VSC).
2. The CIM class name, which is also the SQL table name, of an IBR plant component.
3. The *mRID* of an IBR plant component. This will be a foreign key into the table named in column 2.

To use this table, query for matching the *mRID* of the IBR VSC in column 1. The result may include several rows identifying the balance-of-plant components in columns 2 and 3. This data won’t be in the raw files but should be created in the process of connecting the IBR plant model to the EMT network model.

Chapter 5: IEEE Standards for EMT Modeling

This chapter's objective is to propose a method of ensuring EMT model portability that NERC may suggest as a method of meeting Reliability Standard requirements. Furthermore, the time and cost of implementation should be minimized. IEC and IEEE standards can provide documents that NERC might adopt, or at least reference (e.g., IEEE 2800-2022 for BPS-connected IBRs). A CIM that is fully compliant to IEC standards can satisfy the portability objective but will have a high time and cost of implementation in North America. IEC standards are developed with representation by country while IEEE standards are developed by individuals and organizations. Because NERC members are concentrated in two countries, an IEEE standard development may be easier to initiate and recruit participation. This chapter proposes a simplified CIM coupled with open-source software that would provide examples and starting points for implementers and users.

The IEEE provides a relatively new mechanism for open-source projects in which software may be developed and made available as part of the standards development process.[68] The Apache 2.0 license option allows for commercial adoption without having to license derivative works (i.e., commercial products) under the open-source terms. Therefore, the IEEE open-source platform provides an alternative to the fully IEC-compliant CIM for the rapid development and adoption of practical EMT model repositories. The time frame could range from two to four years, especially if leveraging data formats that are already widely used in North America. Two separate projects are proposed. Both would be Tier 4 IEEE Open-Source projects (i.e., developed as part of companion IEEE Standards rather than standalone software).

1. <u>IEEE Recommended Practice for Electromagnetic Transient Model Interoperability for Electric Power Transmission Systems.</u>[69] A recommended practice has "should do" strength in the ecosystem of IEEE Standards. The scope would include methods of using and extending the new EMT model repository but exclude PSPDT models and distribution systems. It would refer to the next project on real-code modeling. This project will also recommend a fully IEC-compliant CIM implementation, as discussed in **Appendix A:**, for users in position to follow that approach. This will be done by the P3743 Working Group.

2. <u>IEEE Standard for Interfacing Controls and Protection to Power System Simulation Tools.</u>[70] A standard has "shall do" strength in the ecosystem of IEEE Standards. It could have been proposed as a "should do" recommended practice, but upon the IEEE sponsor committee request, it was initiated as a "shall do" standard. Building on [14], this project could formalize inputs and outputs as in **Figure 2.3**. It could generalize the DLL specification for more use cases (e.g., time-series power flow simulations on distribution systems, non-IBR devices, run-time configuration). It could also solicit support from the IEEE PES PSDP Committee because the DLL specification supports PSPDT tools. Alternatives to the DLL specification exist for real-code and other code modeling,[71] but they are more suited for research or development work than for electric utility applications. This standard development project will begin later in 2025 [3]. This will be done by the P3597 Working Group.

A rough outline of the process for the first item, addressing EMT model interoperability, follows:

---

[68] Near the end of 2024, the Working Group for IEEE P1729, Recommended Practice for Electric Power Distribution System Analysis, added an open-source component for maintaining the IEEE Distribution Test Feeders, which had been maintained since 1991 on a variety of web sites and publications. The new open-source repository will formalize the maintenance of these electronic artifacts, including documentation and version control [29]. P1729 has chosen the Berkeley Software Distribution 3-clause license, but the Apache 2.0 license could also be chosen.

[69] This PAR was submitted to IEEE, as project P3743, on July 14, 2025. It was approved at a meeting September 9-10, 2025.

[70] This PAR was approved by IEEE, as project P3597, on May 7, 2025.

[71] MATLAB/Simulink plays an important role in vendor control code verification. Languages like VHDL-AMS (IEEE 1076), Verilog-A and MAST have been used in other domains, e.g., automotive, mixed-signal circuits. The functional mockup interface (FMI), a Modelica Association project, has over two hundred supporting tools ("Functional Mock-up Interface." Accessed: Mar. 16, 2025. [Online]. Available: https://fmi-standard.org/). At least two of those FMI-supporting tools are used by electric utilities for planning and design work, i.e., EMTP® and PowerFactory. Although the DLL interface has more vendor support within the utility industry, FMI could become a viable alternative with more features. The Standard should compare FMI to the DLL interface and identify situations where FMI is applicable.

1. Write an IEEE project authorization request (PAR), including the open-source component, to develop a recommended practice under sponsorship of the IEEE PES Analytical Methods in Power Systems (AMPS) Committee. AMPS is responsible for EMT within the IEEE PES Technical Committee structure. This begins the IEEE process with in-person and virtual meetings, document and software development, balloting and comment resolution, etc.
2. Recruit about 20 working group members from various software tool vendors, consultants, utilities, and other stakeholders. The NERC EMT Modeling Working Group (EMTWG) and the Oak Ridge National Laboratory's annual EMT workshop offer some opportunities for recruitment. The annual CIM User's Group meetings also provide opportunities for recruitment. Contributors and participants need to execute contributor license agreements (CLA) with IEEE. The Apache 2.0 License and CLA help contributors preserve their intellectual property rights.
3. Reference as normative the open-source CIM UML, publicly available at [11].
4. Reference as informative the published DLL specification [14] with a note that the standardized version will be considered normative when available [3]. The membership of this working group and the P3597 working group would overlap. P3597 may not necessarily address all IBR plant parameters that are relevant to EMT studies. If not, this P3743 working group would address them as part of the EMT-IBR profile. Conversely, this P3743 working group is concerned with the complete EMT network model, not just the IBR plants.
5. Reference informative CIM standards for balanced (power flow, short-circuit) network models[72] and PSPDT models, including the WECC generic models[73]. This does not entail adoption of all CIM features, nor the adoption of graph database methods typically used to implement CIM. Use of these CIM standards will minimize effort and duplication of creating new parameter and class names. It may also reduce the effort in adopting more CIM features later.
6. Factor CIM attributes into table definitions that closely match the layout of existing TFFs that are already widely used in North America. These TFFs, sometimes called "raw files," predate the CIM by several decades. Instead of NoSQL databases typically used for full CIM implementations, choose more familiar delimited text file, relational database with SQL, or JSON formats[74]. Compared to the XML files used in CIM, the JSON files are easier to work with. Many programming languages, scripting languages, and open-source tools support JSON. This effort would produce a JSON Schema file, which defines the name, type, format, range, default value, description, and relationships for all attributes in the schema. Documentation, diagrams, and code files are built from the JSON Schema file using appropriate tools, of which there are many options. JSON example files may be validated against the JSON Schema. If the working group chooses a relational database instead of JSON, they will produce data definition SQL. In either case, supporting UML will be produced.
7. Add the essential features of transformer saturation, IBR plants, and DLL interfaces as described in **Chapter 4:**. These could be added to the JSON/SQL schema from item 6, not directly to the CIM UML.[75]
8. The complementary IEEE open-source repository would be populated with test cases and file converters between publicly available CIM input files, and input raw files, and the new EMT schema developed in items 6 and 7. Converters from the schema to open-source EMT tools would also be provided. Working group members may add more examples, e.g., automated scripting of simulation cases and model validations.

[72] IEC 61970-301 Consolidated Version: Energy management system application program interface (EMS-API) - Part 301: Common information model (CIM) base, 2020. Accessed: Apr. 22, 2025. [Online]. Available: https://webstore.iec.ch/en/publication/74467

[73] IEC 61970-302:2024: Energy management system application program interface (EMS-API) - Part 302: Common information model (CIM) dynamics, 2024. Accessed: Apr. 22, 2025. [Online]. Available: https://webstore.iec.ch/en/publication/68152

[74] "JSON Schema." Accessed: Mar. 13, 2025. [Online]. Available: https://json-schema.org/

[75] It may be desirable to incorporate other selected CIM ideas (e.g., bus-branch and node-breaker topology, bus naming, unique identifiers, SI units). Line constants input may also be developed as future extensions. However, these extensions should be deferred to encourage rapid development and adoption of the core schema.

9. Other software vendors, consultants, utilities, and stakeholders would implement their own import and export converters using the EMT repository schema. The published JSON/SQL Schema, documentation and open-source license would allow this activity to occur safely.

The timeline for completion is about three years, which includes one year for the process of IEEE balloting and comment resolution. When completed, the IEEE standard(s) could be adopted by NERC members.

Chapter 6: Revision History

#	Date	Comments
1.0	11/5/2025	Draft for RSTC Review in December 2025

Appendix A: Full CIM Alternative

In this appendix, full CIM refers to implementation in an object-oriented database, including full compliance to the IEC standards and their support for real-time applications. A full CIM-based framework with automation could provide benefits to EMT IBR studies, but with significant future work:

- A CIM-based framework can support automated IBR model validation using scripts and SMIB test systems. The support for different test systems and EMT tools should be broadened. Scripting and test systems are also supported in the CIM-inspired EMT-IBR profile.
- Interoperability tests for PSPDT, SCKT, and EMT models could be sponsored by the CIM user's group. These interop tests have been essential in fixing gaps and errors in the CIM UML, and in promoting or enabling acceptance by utilities and tool vendors. These tests are also included in the EMT-IBR profile.
- Adopt the standardization of asset-based information for transformers and transmission lines. Distribution utilities have already done much of this work through asset management and geographic information systems, which reduces the time to build and validate network models for distribution planning and DER interconnection studies.
- Adoption of node-breaker modeling for EMT and protection system design. This enables more accurate simulation of fault events and protection system response. The bus-branch model, currently used for many PF, PSPDT, and SCKT models, only represents the system state before and after contingencies. Node-breaker is also supported in the EMT-IBR profile.
- CIM fragments for IBR balance-of-plant models, shown in **Figure 2.2**, could help regularize the treatment of network subsystems. IEC Technical Committee 57 Working Group 13 has worked sporadically on the concept of fragments but has not completed it. IBR plant models provide another use case motivating the completion of this work on CIM fragments. This is addressed more simply in the EMT-IBR profile.
- Formalizing the DLL interface as an IEEE Standard under P3597 will foster its adoption in CIM by reference, along with other benefits of standardization. This is preferable to modeling functional logic in CIM. The DLL interface formalization is also part of the EMT-IBR profile.

It is technically feasible to implement an EMT model repository using the full CIM as described above, but significant economic and organizational barriers exist:

- The full CIM has not been widely adopted in North America, so it does not have the same critical mass as in Europe, where a legislative mandate drives model exchange for planning and operations. A relatively small number of North American transmission and distribution utilities have adopted the CIM for other use cases, (e.g., asset management, meter data management). For transmission system analysis, a tool-specific TFF, which predates the CIM by several decades, has already achieved critical mass.
- Intellectual property conflicts have developed between the IEC (standards-making body) and the CIM User's Group (custodians of the data schema in UML, which EPRI originally contributed to the IEC). This increases the cost of development, with duplicate efforts by the organizations. It also creates vendor uncertainty about whether it is safe to use standard code components to implement the CIM. This concern is partly mitigated by the CIM user's group having published the CIM UML under an open-source license, per agreement with the IEC. The license terms are Apache 2.0, which allows for commercial use and derivative works under different licensing terms than open source.
- The cost of relevant IEC standards is higher than IEEE standards. For example, IEEE 2800 costs about $167 for the portable document (PDF) file. In the DER space, a complete bundle of IEEE 1547 costs about $500, and the two essential ("shall do") parts cost about $367. The IEC standards for CIM Dynamics comprise about six parts, at a cost of 2465 Swiss francs or about $2,700. In practice, it's also necessary to join the CIM user's

group for access to the most current UML. This price differential can be a barrier for smaller organizations, like new tool vendors.

The most important advantage of the full CIM alternative is that it would support use cases beyond EMT model maintenance, but that is a higher-level organizational decision.

Appendix B: Extended Raw File Alternative

A TFF, which predates the CIM by several decades, has already reached critical mass in North America. The TFF is sometimes referred to as "raw files." Recently, the TFF vendor adopted an "extended raw file format" based on JSON. During March and April 2025, discussions with the TFF vendor ended with its decision not to contribute resources or file schema to an IEEE standards project. This decision led to the formulation of a lightweight CIM proposal as presented in **Chapter 5:**.

The TFF-based work plan would have been:

- Obtain TFF vendor buy-in, including permission to document TFF in open-source electronic format, which a new working group would then maintain and extend separately from the original software product(s). This buy-in would save development time and increase acceptance of the new EMT network model. The TFF vendor would also participate in this project.[76]
- The TFF vendor was invited to execute CLAs with IEEE [44]. The Apache 2.0 License and CLA help contributors preserve their intellectual property rights.
- The new schema would have been documented from an existing JSON version of TFF. Compared to the XML files used in CIM, the JSON files are easier to work with. Many programming languages, scripting languages, and open-source tools support JSON.
- Technical work would have begun with the automated generation of JSON Schema from the TFF. This roughly corresponds to the data modeling already in CIM with Dynamics. Only the essential extensions for transformer saturation, IBR plants, and DLL interfaces would have been added as discussed in Chapter 4:. Consultants and tool vendors might have contributed file format converters and test cases.
- The effort would have published a JSON Schema file, which defines the name, type, format, range, default value, description, and relationships for all attributes in the schema. Documentation, diagrams, and code files would have been built from the JSON Schema file using appropriate tools, of which there are many options. JSON example files might have been validated against the JSON Schema.
- The IEEE working group, collaborating with or including the TFF vendor, would have maintained compatibility between the EMT repository schema and the TFF software product. This would have allowed both schemas to evolve along different paths at different paces. The lifecycle of an IEEE Standards project is 10 years, so the TFF product would be expected to update more frequently. The standard and JSON Schema would have been written to allow for extensions and versions. This would have simplified the process of maintaining the standard on 10-year cycles. If the standard and TFF software product fell out of synchronism, two options could have closed the gaps. First, customer-driven support for legacy file export might have been included in the TFF software vendor's product. Second, if that did not occur, then interested parties might have written their own version translators using the published JSON Schema.
- Other software vendors, consultants, utilities, and stakeholders would have implemented their own import and export converters using the EMT repository schema. The published JSON Schema, documentation and open-source license would have allowed this activity to occur safely.

[76] Working group members may expect to spend up to eight hours each month (average) on virtual meetings and off-line contributions for up to four years. The TFF vendor might have assigned one (or more) employees join the working group. It was recommended, but not required, that participants be individual members of the IEEE Standards Association, so they can vote on the standard during balloting. This project will follow the individual process for IEEE standards, which means that working group members agree to exercise their own professional judgement. Members declare their affiliations, either an employer or a funding sponsor. In this project, the TFF vendor's direct interests might have been met via the Contributor License Agreements (IEEE SA Open, "Apache 2.0 CLA · IEEE," GitLab. Accessed: Mar. 13, 2025. [Online]. Available: https://opensource.ieee.org/community/cla/apache).

Some of this TFF interface work already occurs ad hoc by various industry stakeholders. Without standards, this ad hoc work is duplicative and prone to version mismatch or other errors. Ad hoc extensions for the DLL interface, IBR plant parameters, transformer saturation, etc., will not be portable, especially if they are not required in the PF and PSPDT use cases. Standardization of the TFF would have addressed all these barriers for the EMT stakeholders. Standardization could have led to demand for new products, benefiting the TFF vendor and others.

The EMT-IBR profile as proposed in **Chapter 5:** will meet all objectives that the TFF-based project might have achieved. TFF vendor participation in the EMT-IBR profile would still be helpful but not required. The EMT-IBR profile also provides a path for interested stakeholders to adopt the full CIM in the future.

Appendix C: Experience with DLL Interface

Test cases and support code for the DLL interface have been published to an open-source repository. The code builds 32-bit and 64-bit versions of three example DLLs, described later. Standalone test harnesses have been provided to run these DLL models and plot the outputs (i.e., without any connection to a power system). A wrapper library has been provided to handle some details of parameter alignment, memory management, etc. The experience has been successful, with a few observations:

- Struct alignment was uncertain because the DLL interface currently supports many types of parameters and signals, not just double-precision real numbers. Comments from this experience led to some clarifications in Cigre JWG B4.82/IEEE Guidelines for use of real-code in EMT models for HVDC, FACTs and inverter based generators in power systems analysis.
- The DLL interface requires that the time step be fixed and unchangeable before run time. The ATP implementation mitigates this by only calling the DLL when it needs to catch up with ATP's time, and as often as needed. This means the DLL time and ATP time can fall out of step. This is inherent in the DLL interface specification.
- The signal and parameter names are also fixed before run time.
    - The second example presented here takes just a single parameter, the name of a JSON configuration file with many trained data-driven coefficients, the preferred time step, etc. The signal connections must be known ahead of time per the DLL interface specification, which defeats flexibility offered by the JSON configuration file.
    - The DLL specification does not specify signal names or essential IBR plant parameters. This could vary with different model providers. Additional detail in the specification could reduce these friction points.

The ATP examples have been posted to an open-source repository[77]. To run these examples, users need an ATP license. The ATP interface has limitations due to its decades-old design and implementation, but it works well enough to demonstrate use of the DLL models in EMT simulation, with no software licensing cost required. ATP's MODELS feature wraps the DLL function calls. It's necessary to edit and compile at least two source files, and link them to the main solver, whenever a new DLL is to be called from ATP. Kestrel EMT examples from the 2025.10 version of the software are also shown. These work natively with the IEEE/CIGRE interface DLLs, with no additional source code compilation required.

When it supports the DLL interface, NREL's ParaEMT simulator's repo[78] could also host these three examples. This option is attractive because ATP licensing terms, while zero cost, complicate the distribution of ATP examples just as if ATP were a commercial product. One potentially important limitation of ParaEMT is that it offers pi-section line models but not distributed-parameter (Bergeron) line models. Bergeron and other distributed-parameter line models tend to be more efficient and robust in large EMT networks. Fortunately, the basic Bergeron models are not difficult to implement in code.

[77] T. McDermott, temcdrm/emthub. (Sept. 25, 2024). Python. Accessed: Sept. 25, 2024. [Online]. Available: https://github.com/temcdrm/emthub

[78] NREL/ParaEMT_public. (Apr. 03, 2025). Python. National Renewable Energy Laboratory. Accessed: Apr. 10, 2025. [Online]. Available: https://github.com/NREL/ParaEMT_public

Excitation System Example

The excitation system example, *scrx9*, comes with the IEEE Cigre JWG B4.82 task force report. A test system was obtained from a Modelica implementation of this example[79], and later implemented in ATPDraw[80] (**Figure C.1**). A three-phase fault occurs from 2.0 to 2.15 seconds, about halfway into the SMIB grid equivalent. **Figure C.2** and **Figure C.3** show correct performance of this 32-bit DLL model in ATP.

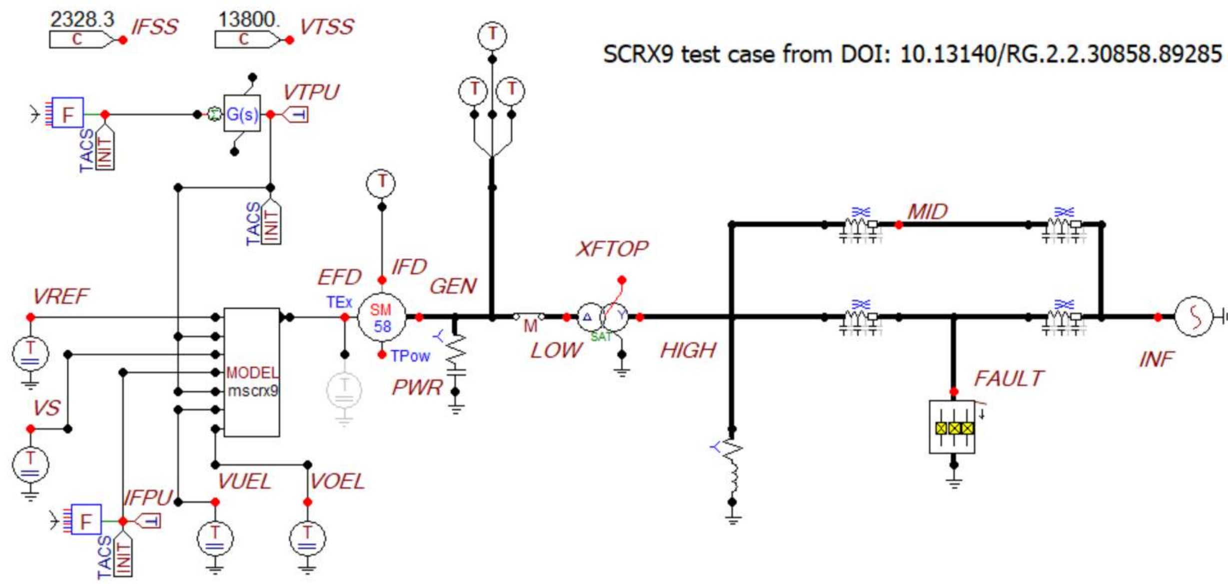


Figure C.1: Circuit of the Static Exciter Connected to a Machine and SMIB Network

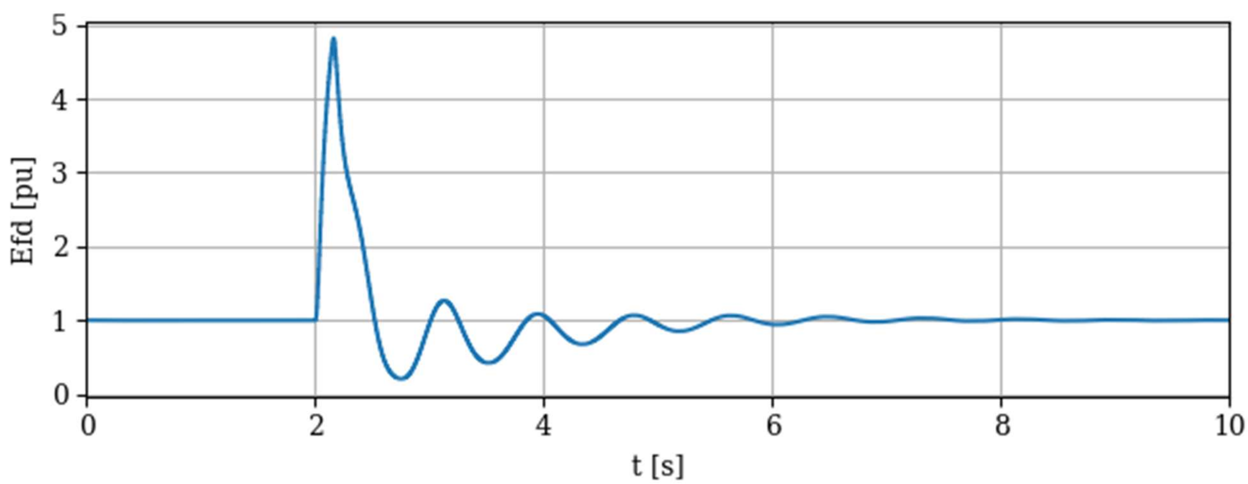


Figure C.2: Simulated Field Input Voltage [pu] Normalized to the Initial Condition

[79] H. Chang and L. Vanfretti, "Integrating the IEEE/CIGRE DLL Modeling Standard to Use 'Real Code' Models for Power System Analysis in Modelica Tools," Modelica Conferences, pp. 72–79, 2024, doi: 10.3384/ecp20772.

[80] H. Chang, "embeddedhao/OpenIPSL at cigre_dll_pr," GitHub. Accessed: Apr. 08, 2025. [Online]. Available: https://github.com/embeddedhao/OpenIPSL

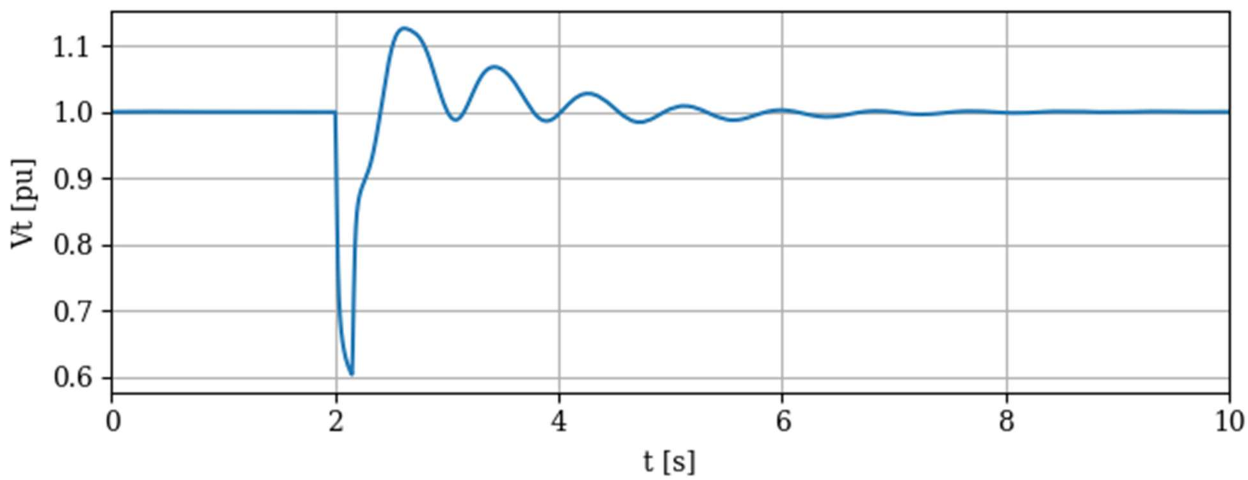


Figure C.3: Terminal Voltage [pu] Dips to 0.60387 vs. 0.57912 Without the Excitation System

Data-Driven IBR Model Example

This example derives from a U. S. Department of Energy (DOE)-funded project on grid-forming inverter control[81]. That project included the development and training of data-driven IBR models, referred to as Hammerstein-Wiener Photovoltaic (HWPV) models. The circuit in **Figure C.4** was used to create training data for a model of the 100-kW Consortium for Electric Reliability Technology Solutions microgrid inverter. **Figure C.5** connects the trained HWPV DLL model to a grid. The control signals and *dq0* transformations were implemented in ATP's legacy Transient Analysis of Control Systems (TACS) feature, which is several times faster and more robust than ATP's newer MODELS feature for control systems. **Figure C.6** through **Figure C.11** demonstrate proper performance of the HWPV DLL through a sequence of several disturbances in control signals, solar irradiance, temperature, grid load, and a grid fault. The HWPV model runs about as fast as an average EMT model of the GFM IBR, and many times faster than a switching EMT model of the GFM IBR. A single line-to-ground fault (SLGF) occurs from 9.0 to 10.0 s, during which the oscillations in *dq0* quantities are expected. However, these results were obtained with a balanced model called *bal3n*, not the unbalanced version *unb3*. As newly trained models are added to the HWPV repository[82], the results may change. Under another DOE project, PNNL is currently developing HWPV models of DER for aggregation into bulk power system EMT studies.

[81] Z. Qu, "Autonomous Inverter Controls for Resilient and Secure Grid Operation: Vector Control Design for Grid Forming," DE-EE0009028 Final Report, 2481660, May 2024. doi: 10.2172/2481660.

[82] T. E. McDermott et al., "Deep Learning-enhanced Block-Diagram Modeling of Solar Power Systems," in 2024 9th IEEE Workshop on the Electronic Grid (eGRID), Nov. 2024, pp. 1–6. doi: 10.1109/eGRID62045.2024.10842890.

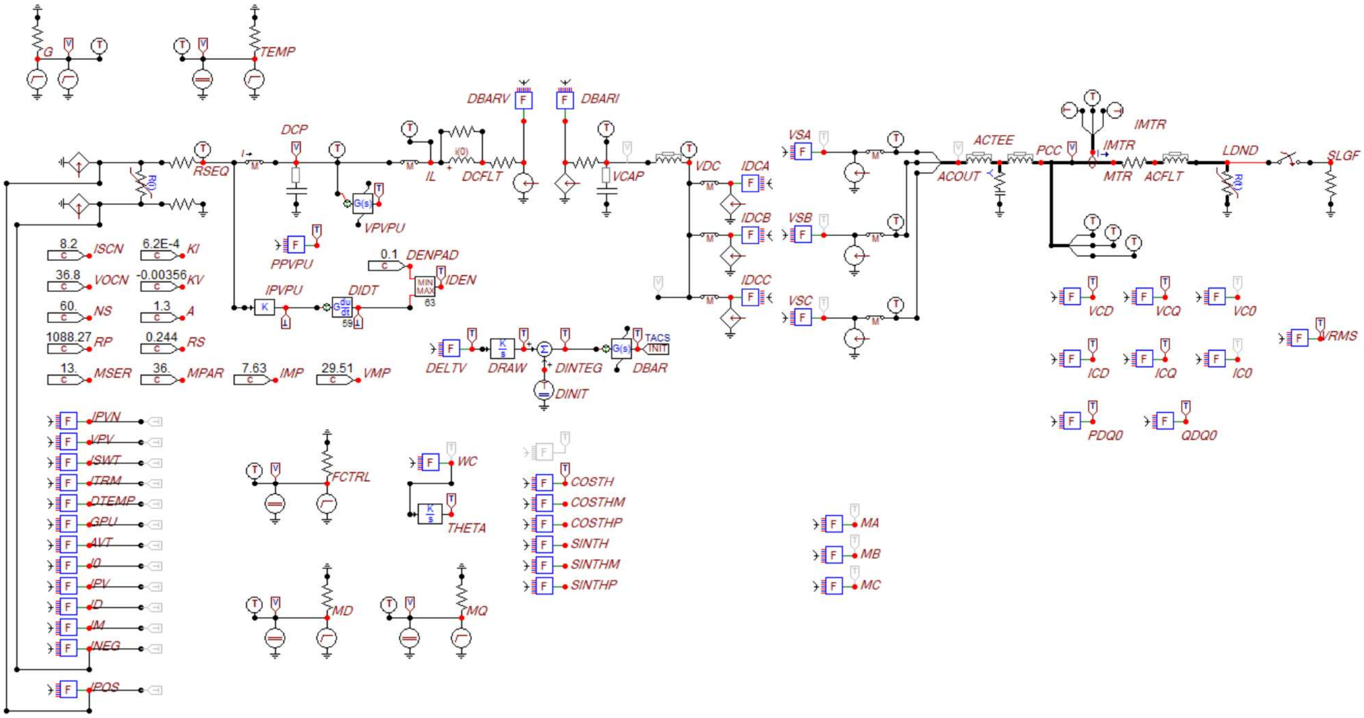


Figure C.4: Circuit of the 100 kW Average Inverter Model Used to Train the Data-Driven Model

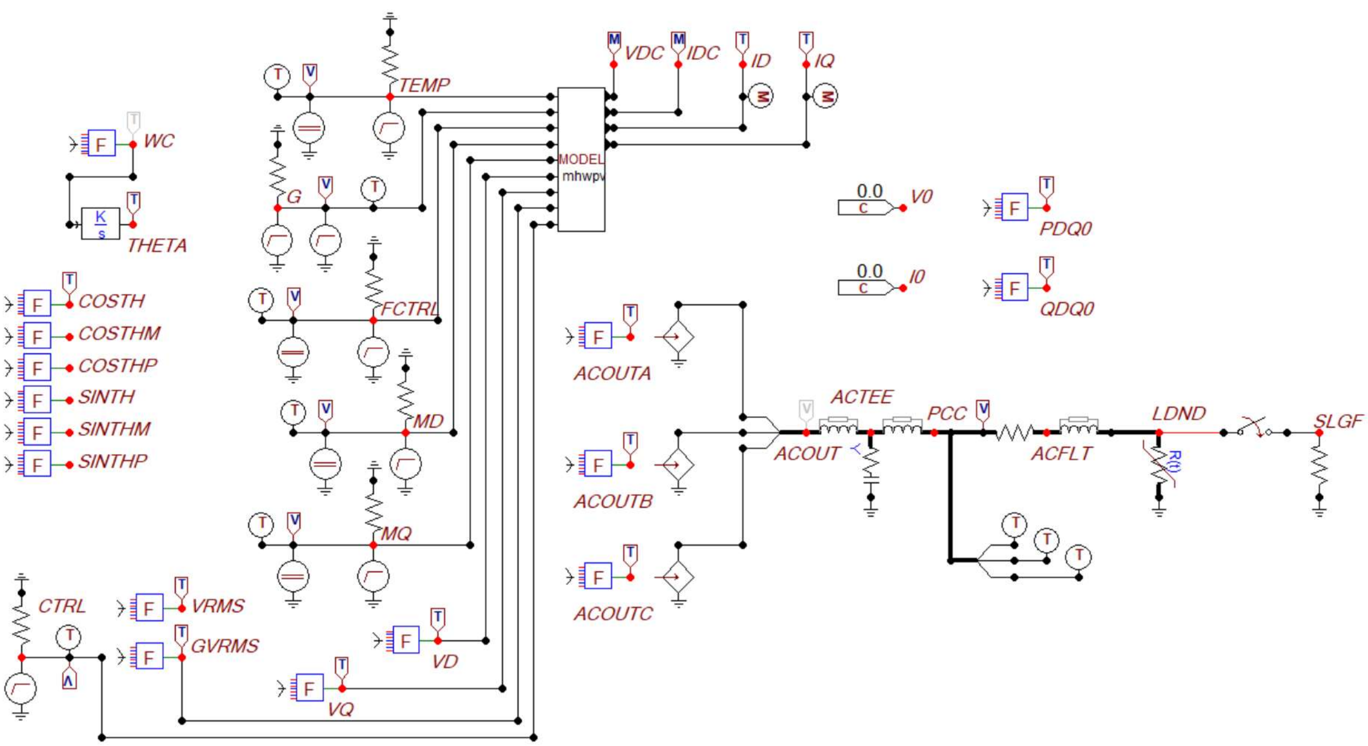


Figure C.5: Circuit of the 100 kW HWPV Model Controlling Current Sources Connected to a Filter and Load (Lower Right), with Control Signals and dq0 Transformations (Left)

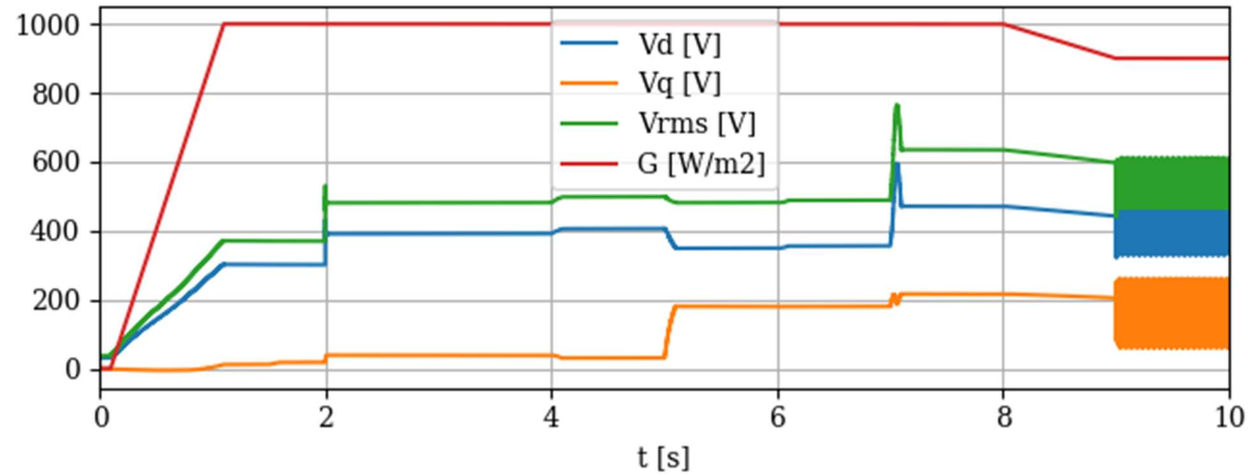


Figure C.6: Simulated Input Voltages and Solar Irradiance with the HWPV Model. G Ramps from 0 to 1000 Over t=0.1 to 1.1 s Then to 900 Over t=8 to 9 s. The Load Resistance Changes from 2.304 to 3.304 ohms at 2.0 s, Representing a Change from 100 kW to 69.7 kW Nominal Load. A Permanent SLGF Occurs on Phase A at 9.0 s.

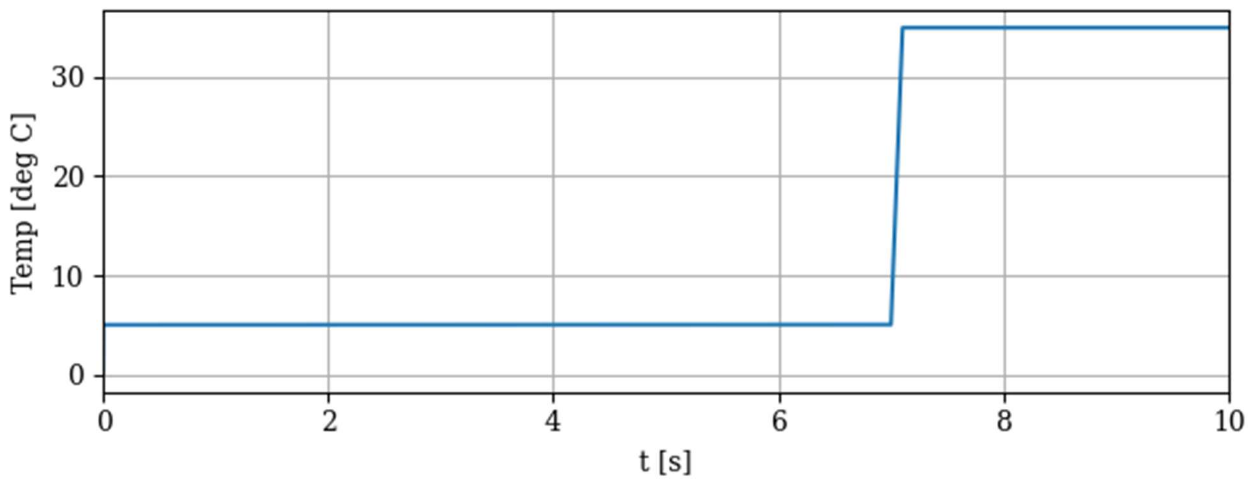


Figure C.7: Simulated Temperature Input Ramps from 5 to 35 over t=7.0 to 7.1 s

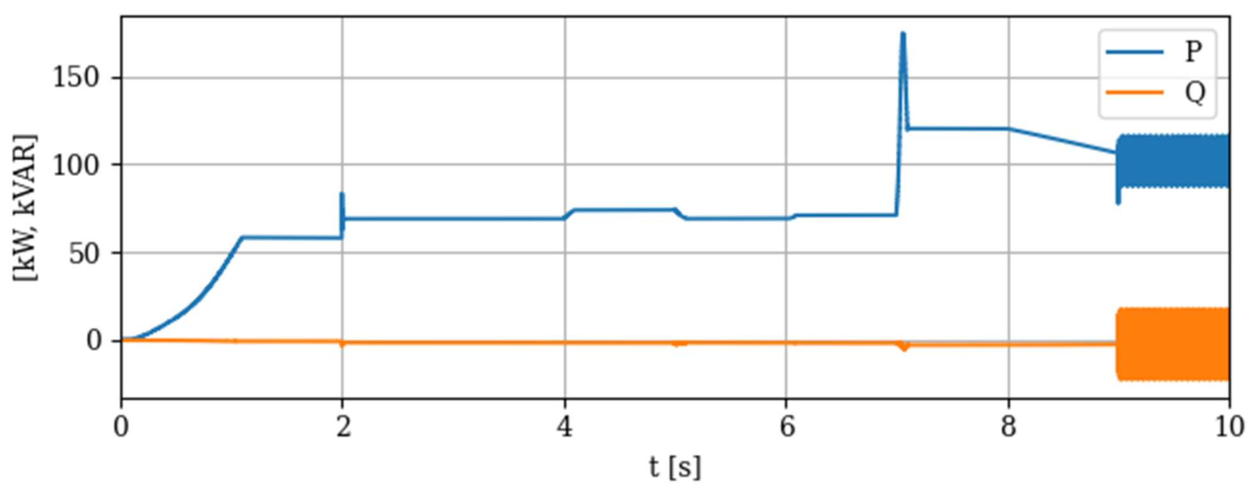


Figure C.8: Simulated Output Active Power and Reactive Power Calculated from dq0 Components

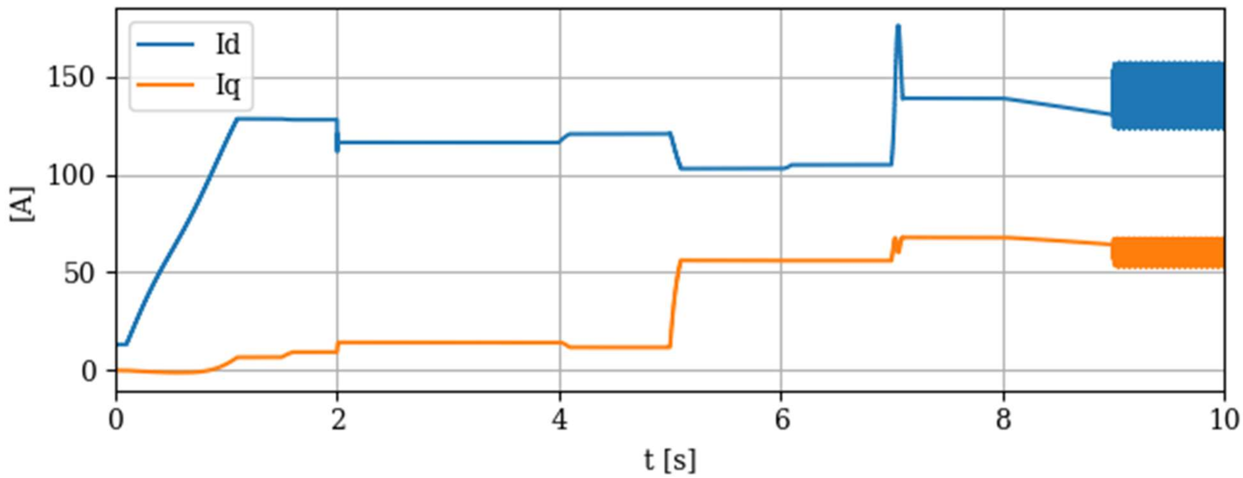


Figure C.9: Simulated Direct and Quadrature Axis Output Currents Injected as Norton Source Currents to the Grid

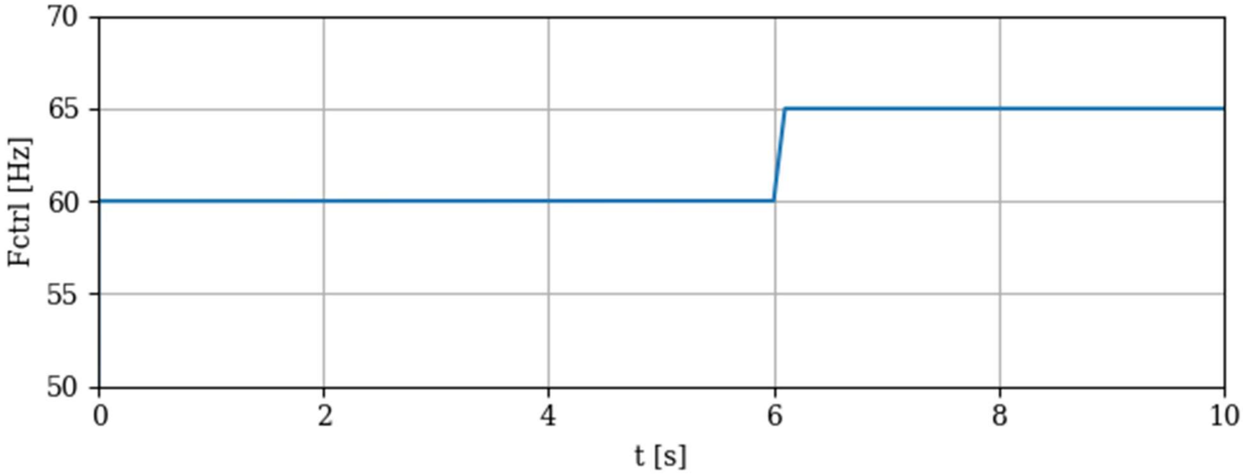


Figure C.10: Simulated Input Control Frequency Ramps from 60 to 65 Over t=6.0 to 6.1 s

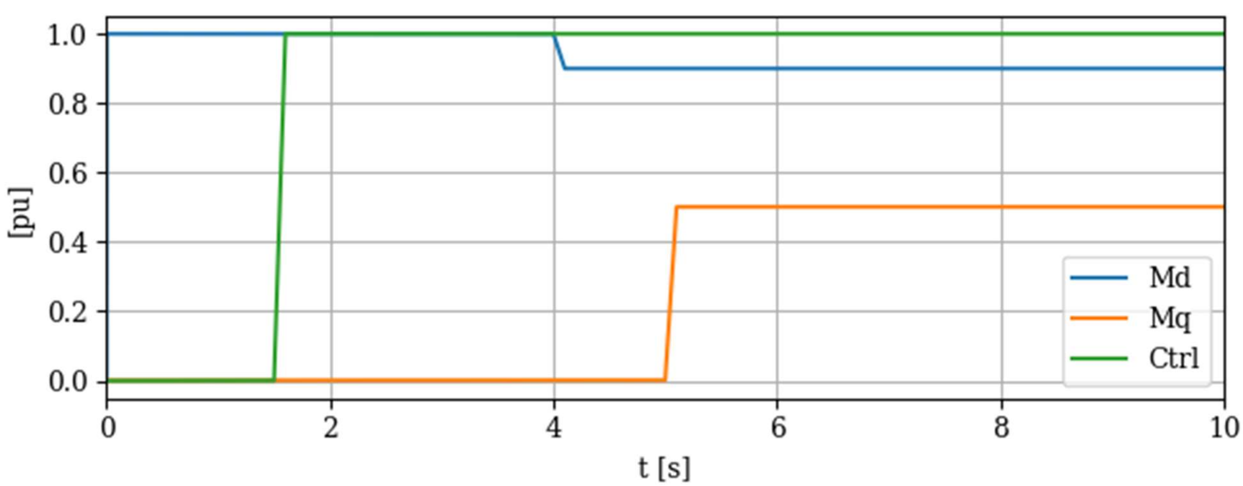


Figure C.11: Simulated Input d-axis and q-axis Control Voltages and inverter operating mode. Md Ramps from 1.0 to 0.9 Over t=4.0 to 4.1 s. Mq Ramps from 0.001 to 0.5 over t=5.0 to 5.1

s. The Inverter Operating Mode, Ctrl, Changes from 0 (startup) to 1 (grid formed) Over t=1.5 to 1.6 s.

Grid-Forming Inverter Model Example (ATP)

EPRI has published an average model[83] of a GFM/GFL IBR [35]. The example in this report uses a DLL code provided to the Cigre JWG B4.82 task force on January 27, 2022, which is earlier than the September 21, 2023 code listing that was published in EPRI's "Code Based Generic Inverter Based Resource Model". **Figure C.12** shows this GFM DLL connected to a SMIB, along with control signals. The model inputs include some filter parameters, which are shown with default values in **Figure C.13**. This topology was not documented in the EPRI's model or elsewhere, so it was verified via email with EPRI.[84] The converter DLL attempts to control voltage and current at the control point, V and I, using controlled voltage sources, E, along with measured pre-filter currents I_L. **Figure C.14** through **Figure C.19** show proper operation of this DLL model through initialization and a SLGF from 2.0 to 2.15 s. Limitations in ATP and MODELS have so far prevented smoother start-up.

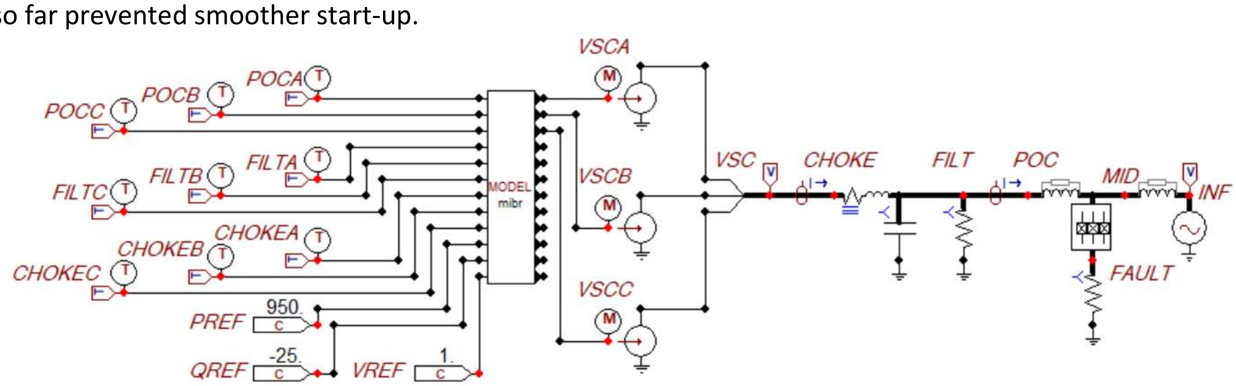


Figure C.12: 1-MW Average Inverter Model DLL Connected to a Filter and SMIB (ATPDraw)

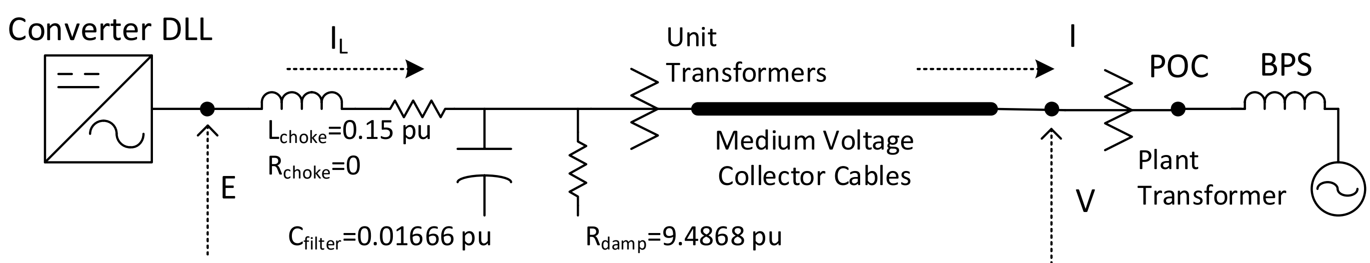


Figure C.13: Output Filter and Plant Configuration. EPRI Recommends the Control Point be Located on the Low Side of Any Plant Transformer, Not the High-Side Point of Connection (POC).

[83] An average model uses controlled sources to represent the IBR. A switching model uses power electronic switches to represent the IBR, which entails a lower time step, and much longer EMT simulation times compared to the average models.

[84] The default damping resistance may be too large for a series connection with the filter capacitance, but it seems too small for a parallel connection. Figure C. now shows a noticeable difference in pre and post filter currents.

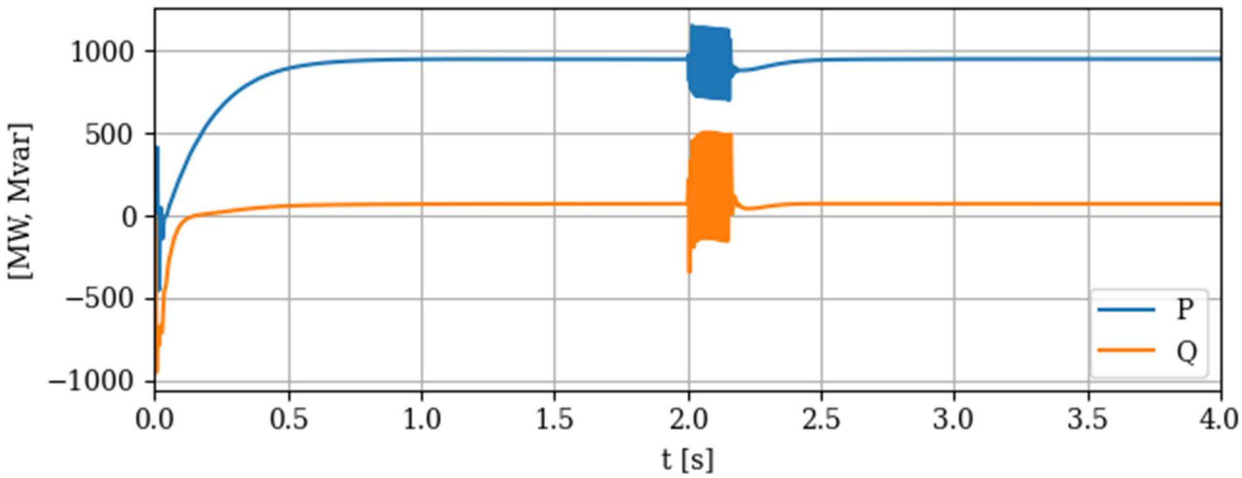


Figure C.14: Simulated IBR Active Power [MW] and Reactive Power [Mvar]

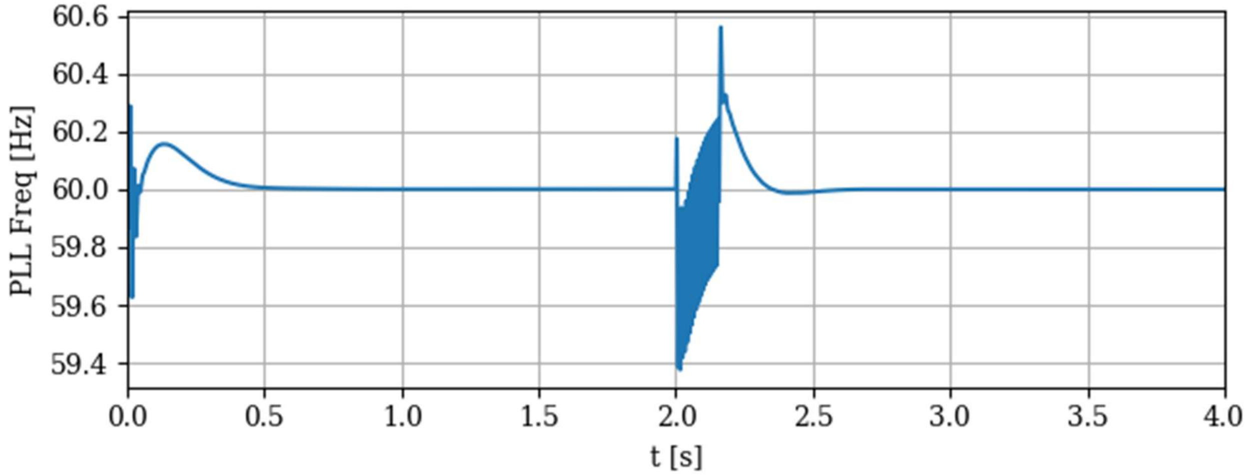


Figure C.15: Simulated IBR Phase-Locked Loop Frequency [Hz]

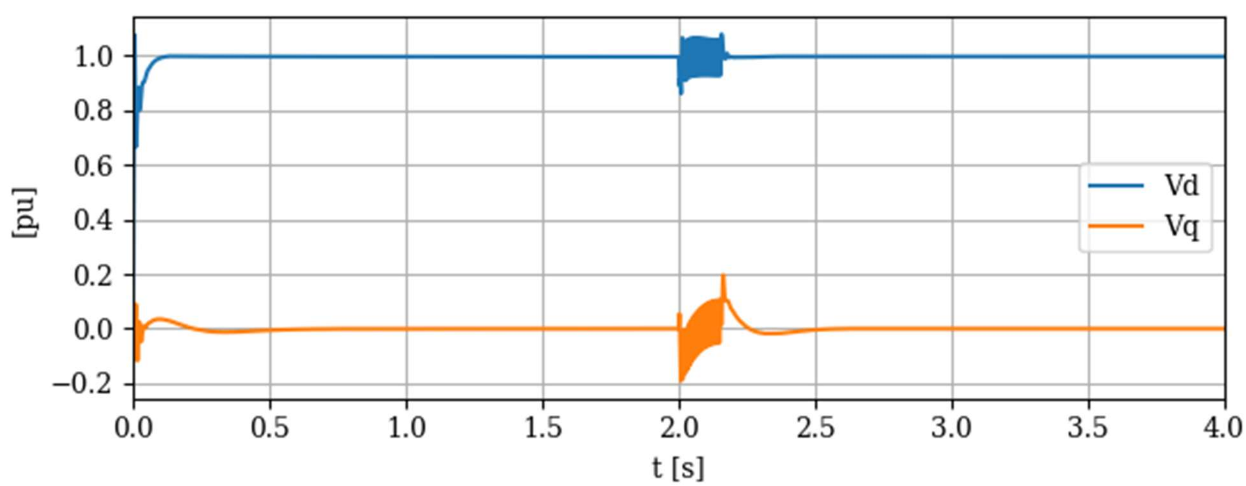


Figure C.16: Simulated IBR d and q Axis Voltages [pu]

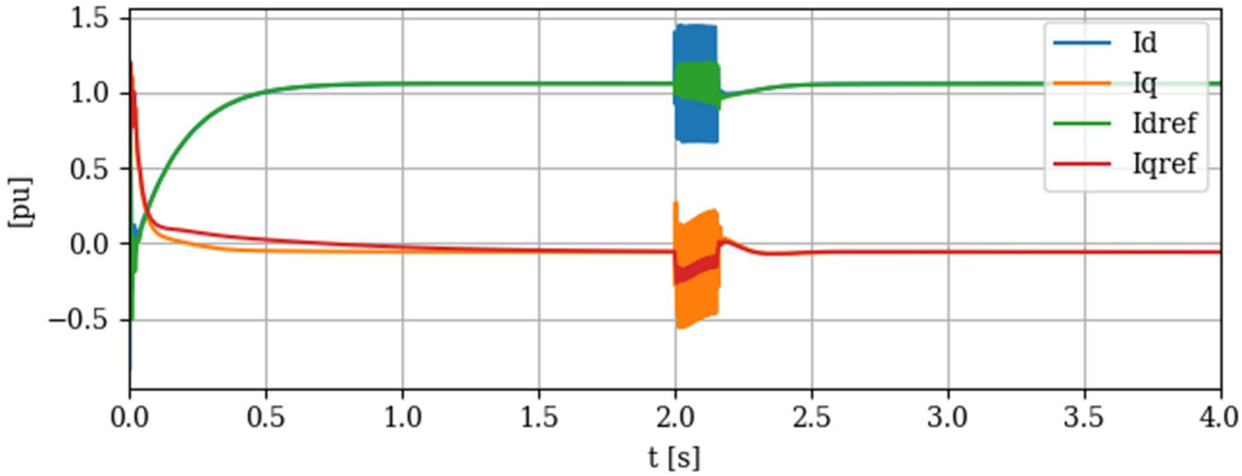


Figure C.17: Simulated IBR d and q Axis Reference Currents and Output Currents [pu]

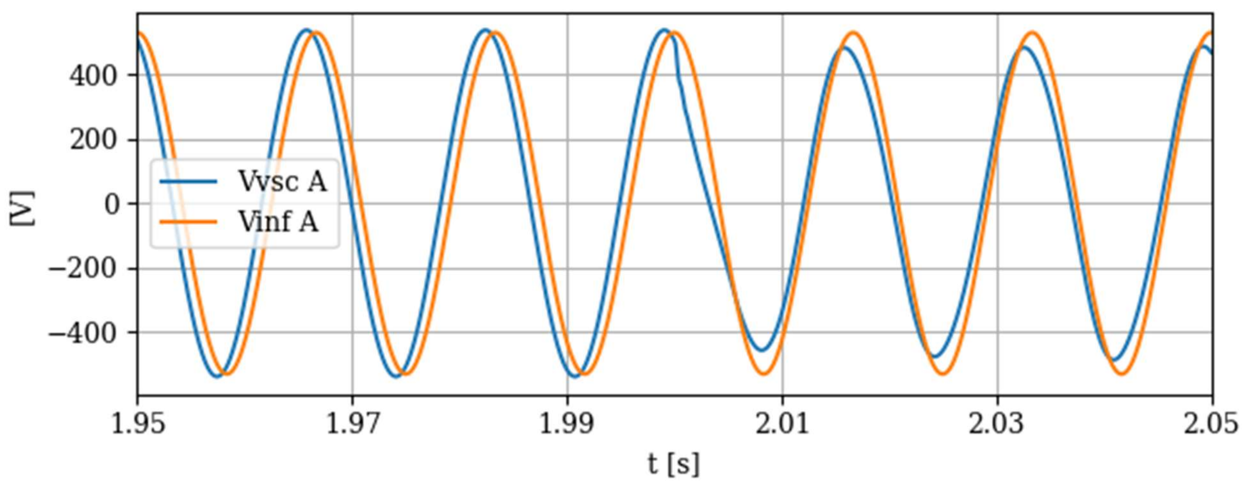


Figure C.18: Simulated IBR VSC and Infinite Bus Voltages on Phase A; Angle Difference Creates Active Power Flow

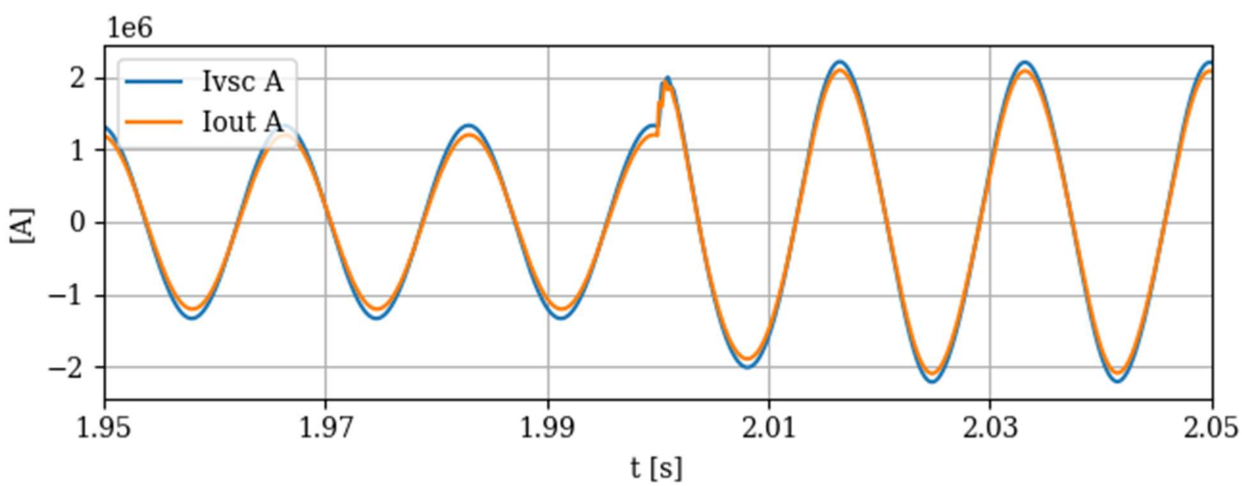


Figure C.19: Simulated IBR VSC Output Current Before and After the Filter Capacitor

Grid-Forming Inverter Model Example (Kestrel EMT)

This example uses the EPRI GFM/GFL IBR average model that was published in [35]. Figure C.20 shows this GFM DLL connected to a SMIB, along with control signals. Just as in the ATP-EMTP example, the converter DLL attempts to control voltage and current at the control point using controlled voltage sources, along with measured pre-filter currents. Figure C.21 through Figure C.23 show proper operation of this DLL model through initialization and a SLGF from 4.0 to 4.15 s.

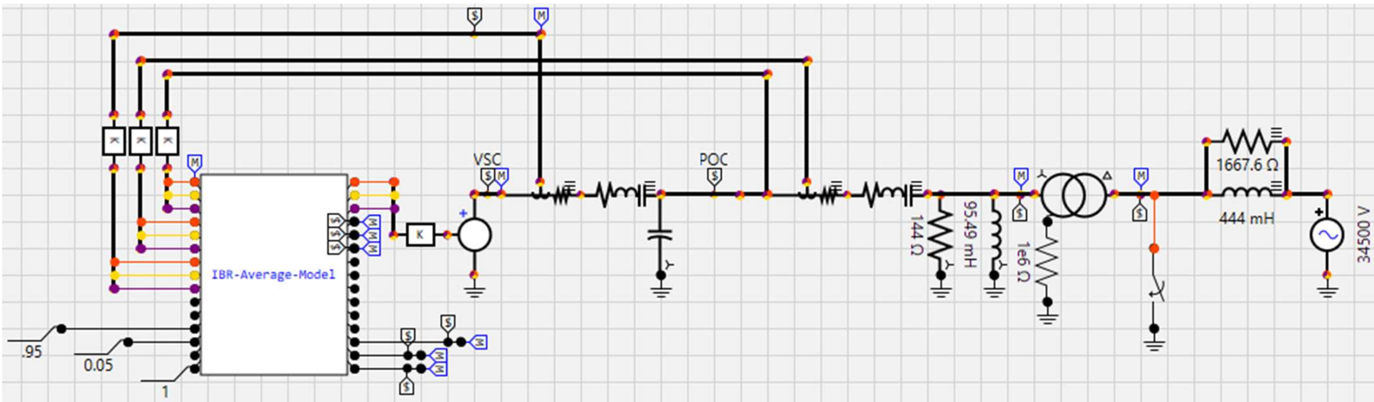


Figure C.20: 1-MW Average Inverter Model DLL Connected to a Filter and SMIB (Kestrel EMT)

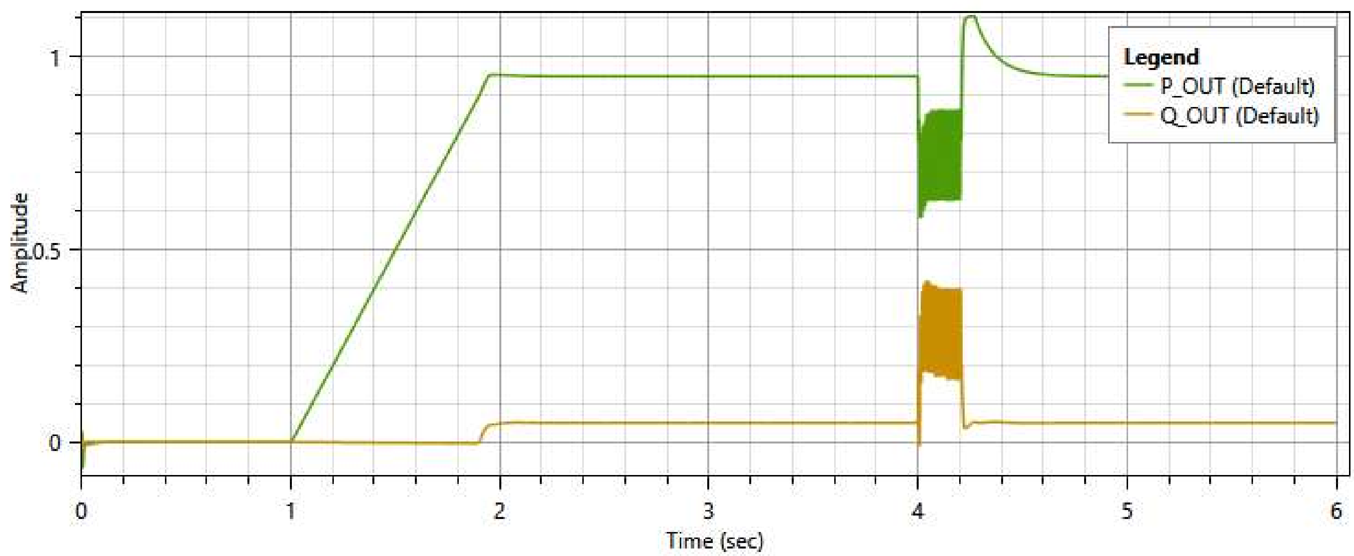


Figure C.21: Simulated IBR Active Power [pu] and Reactive Power [pu].

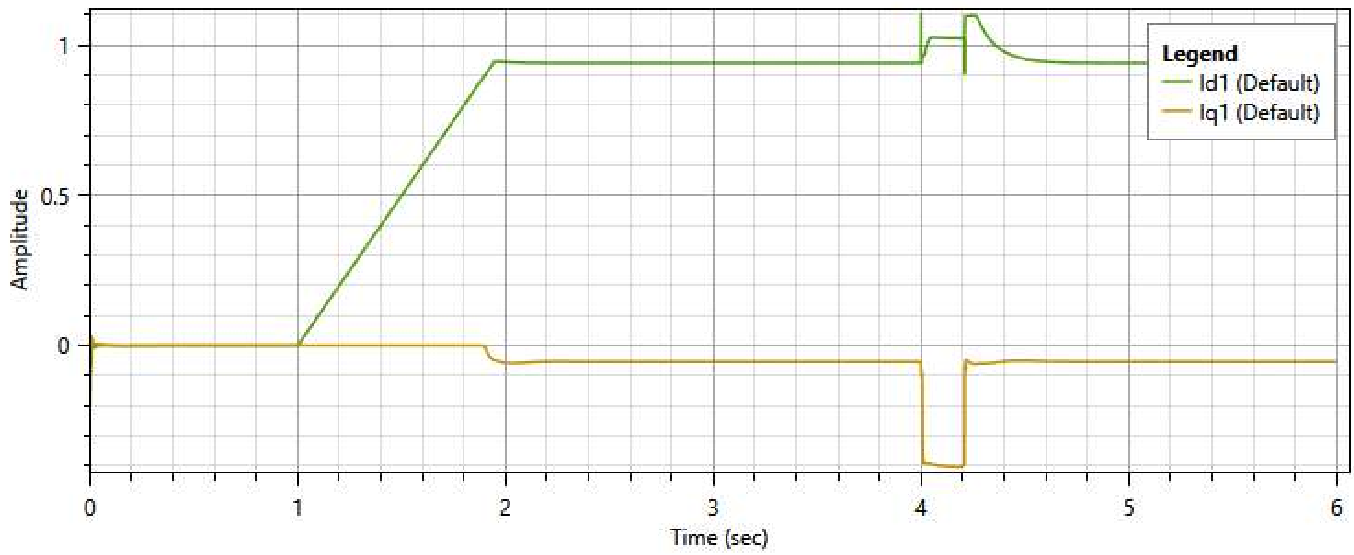


Figure C.22: Simulated IBR d and q Axis Currents [pu]

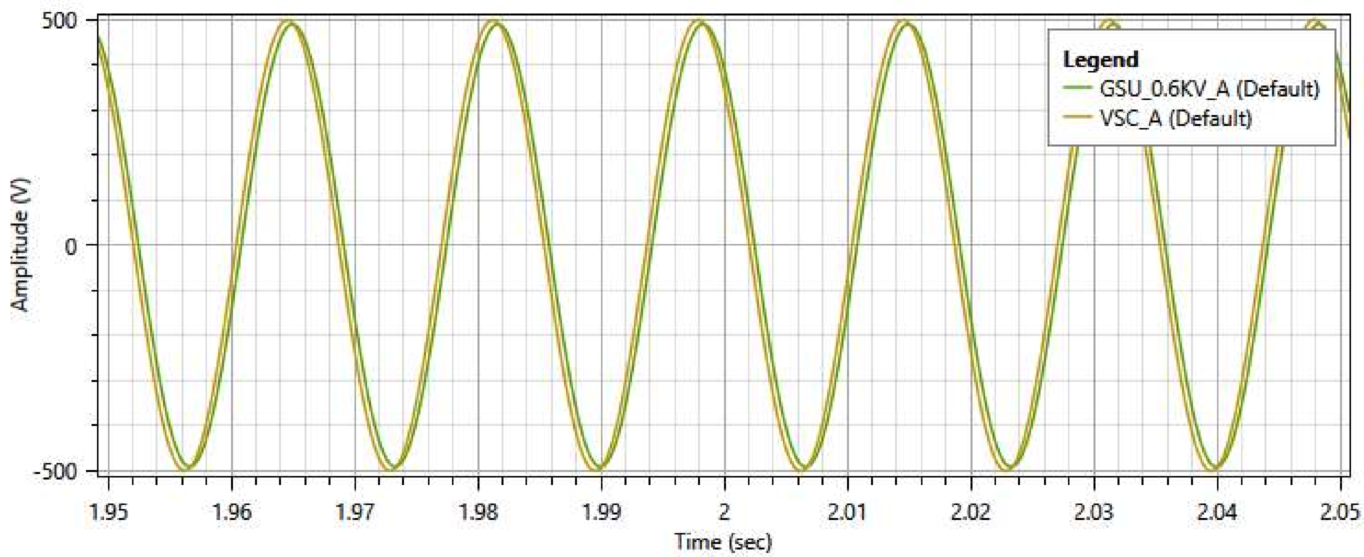


Figure C.23: Simulated IBR VSC and Infinite Bus Voltages on Phase A; Angle Difference Creates Active Power Flow

Appendix D: Abbreviations

List of Abbreviations	
Abbreviations	**Full description of term**
AC	Alternating Current
AMPS	IEEE Analytic Methods for Power Systems
AMS	Analog Mixed Signal
API	Application Program Interface
ATP	Alternative Transients Program
BPS	Bulk Power System
CGMES	Common Grid Model Exchange Specification (of ENTSO-E)
CIM	Common Information Model
CIP	Critical Infrastructure Protection (NERC Standard)
CLA	Contributor License Agreement
COMTRADE	Common Format for Transient Data Exchange for Power Systems
CSV	Comma-Separated Values
CT	Current Transformer
DC	Direct current
DDL	Data Definition Language
DER	Distributed Energy Resource
DLL	Dynamic Link Library
EHV	Extra-high Voltage
EMS	Energy Management System
EMT	Electromagnetic Transient
ENTSO-E	European Network of Transmission System Operators for Electricity
EPRI	Electric Power Research Institute
FACTS	Flexible AC Transmission System
FERC	Federal Energy Regulatory Commission
FMI	Functional Mockup Interface
GFL	Grid-Following Inverter
GFM	Grid-Forming Inverter
GIS	Geographic Information System
GSU	Generator Step-up Unit (Transformer)
HVDC	High-Voltage, Direct Current
HWPV	Hammerstein – Wiener Photovoltaic model
IBR	Inverter Based Resources
IEC	International Electrotechnical Commission
IED	Intelligent Electronic Device
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
MAST	Modeling Language for the Saber™ System Simulator
MOD	Modeling, Data, and Analysis (NERC Standard)

List of Abbreviations	
Abbreviations	**Full description of term**
MODELS	A control system modeling feature of ATP
NoSQL	Not Structured Query Language
NREL	National Renewable Energy Laboratory
PAR	(Standards Development) Project Authorization Request
PC	Planning Coordinator
PDF	Portable Document Format
PF	Power Flow (Analysis Tool)
PNNL	Pacific Northwest National Laboratory
POC	Point of Connection (for BPS connections)
PRC	Protection and Control (NERC Standard)
PSDP	IEEE Power System Dynamic Performance Committee
PSPDT	Positive Sequence, Phasor Domain Transient (Analysis Tool)
PV	Photovoltaic
PWM	Pulse Width Modulation
RMS	Root Mean Square
SCKT	Short-Circuit (Analysis Tool)
SI	International System of Units
SLGF	Single-Line-to-Ground Fault
SMIB	Single-Machine, Infinite Bus
SPARQL	A Web Standard Protocol and Query Language for Graph Databases
SQL	Structured Query Language
SSR	Sub-Synchronous Resonance
SVC	Static Var Compensator
TF	Task Force
TFF	Text File Format
TP	Transmission Planner
UML	Unified Modeling Language
UUID	Universally Unique Identifier
VSC	Voltage-Source Converter
VT	Voltage Transformer
WECC	Western Electricity Coordinating Council
XML	Extensible Markup Language

Appendix E: Contributors

Contributors	
Name	**Organization / Institution**
Daigle, David	Arnada Engineering
Lapointe, Vincent	Opal-RT
McDermott, Thomas	Meltran
Thant, Aung	North American Electric Reliability Corporation